

IS-Switch **ISO8583** **Message Processing Specification**

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DOCUMENT CHANGE HISTORY

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1.0 General concepts about Electronic Data Capture (EDC)

The primary function of an Electronic Data Capture (EDC) terminal is to interface with a transactions acquirer Gateway or also called “Transactional Switch”.

EDC capability allows the terminal to transmit transaction data directly to a specialized “POS End Point”, part of the “Switch” system, which receives the authorizations and financial submission transactions.

Message types processed and/or captured through the EDC Interface include:

- Authorizations
- Sales
- Refunds
- Voids
- Debit and Credit Adjustments
- Debit and Credit File Updates
- Reversals
- Financial Reconciliation (Submission)
- Maintenance

2.0 Reconciling and Closing Batches

Reconciliation of transactions will need to take place to obtain submission for those transactions conducted since the last submission. It is anticipated that such reconciliations will be performed daily and may be triggered automatically or manually for each Acquirer as desired by the terminal owner or Acquirer.

When reconciliation processing has completed successfully, the Merchant may process transactions for the next submission period.

2.1 Batch out of Balance

If the reconciliation totals do not match between the terminal and the server stored totals, the Batch Close response will return a response code informing the POS device to upload all the transactions in the device's batch. As soon as the POS device receives this response, the device will transmit each of the transactions in its batch to "ISSwitch Server". The Server, upon receiving these uploaded transactions will store them in a new batch. This process continues until the POS device completes this cycle by sending another Batch Close request. Once the batch is closed at the terminal, the transactions will be removed from the terminal's memory.

2.2 Close Batch Procedure

The terminal will determine if there is a reversal and/or outstanding transactions (like Offline Authorized Transactions or any kind of Advice Transactions) to be trickle fed (Uploaded to the Switch). If so, it will attempt to establish communications with "ISSwitch Server" and then all outstanding 0220 and 0400 messages must be sent to the host prior to attempting reconciliation. When the *trickle feed*¹ is complete, the reconciliation message may be sent. If there is no reversal and/or outstanding transactions to be trickle-fed, the terminal should proceed with the submission directly.

¹ Trickle Feed. Process where the pending advices and reversals are sent to the Acquiring Switch to be processed

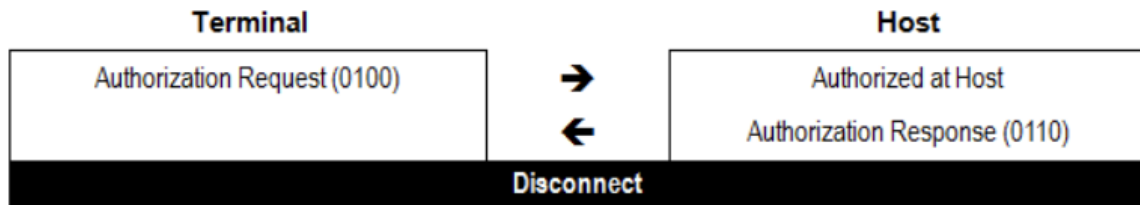
3.0 Transaction Flows

This section shows, in diagrammatic form, the way in which the transaction dialogue is conducted between the terminal and the host.

3.1 Authorizations

3.1.1 Normal Processing

An Authorization Request (0100) message is sent to the host with the request details. The host sends an Authorization Response (0110) message, which represents either an approved transaction, a denial or a voice-referred transaction. If a transaction is referred to voice authorization, and the Issuer approves the transaction, the Issuer provides an approval code to the Merchant. The Merchant normally enters the approval code into the terminal and the voucher is printed then.



3.1.2 Reversal Processing

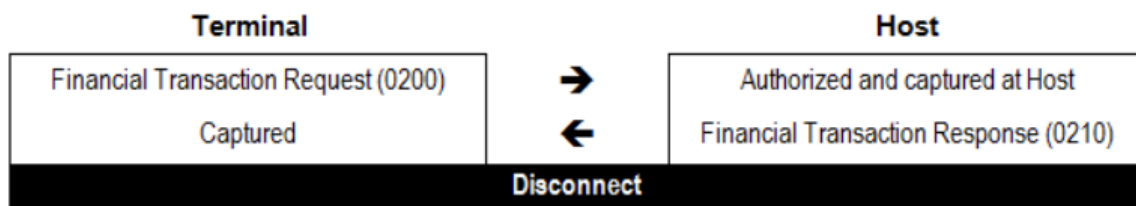
Typically, the terminal will time-out if a response message has not been received within the terminal time-out value. If an Authorization Request (0100) pre-authorization transaction times out, a Reversal Request (0400) message is required.

3.2 Financial Transactions

3.2.1 Normal Processing

The following diagram shows the normal processing flow for financial transactions. A Financial Transaction Request (0200) message is sent to the host, which returns a Financial Transaction Response (0210) message to the terminal.

This general flow applies to transactions, which are approved during the authorization process. Transactions that are referred for voice authorization are not represented in this flow diagram.



3.2.2 Reversal Processing

Typically, the terminal will time-out if a response message has not been received within the terminal time-out value. If an Authorization Request (0100) pre-authorization transaction times out, a Reversal Request (0400) message is required.

3.2.3 Online Voids

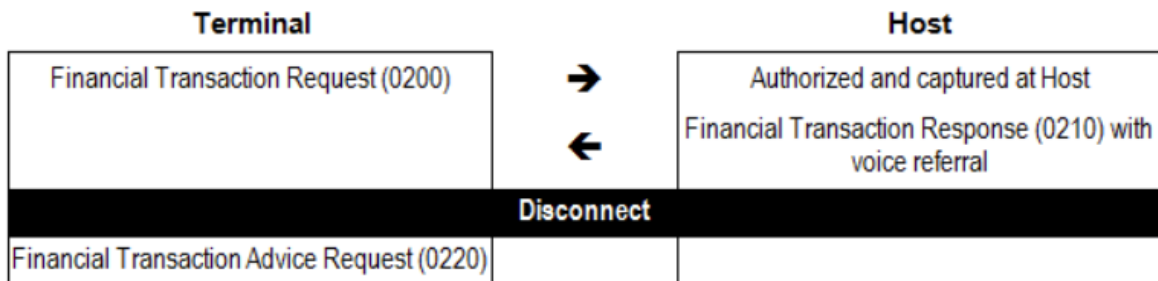
If the terminal attribute allows for an online void, they can be generated for online sales, online refunds, offline sales and offline refunds that have been processed by the Host. Online voids of online sales or offline sales are sent by a Financial Transaction (0200) message with a processing code of "02 XX XX". Online voids of an online refund or offline refund are sent by a Financial Transaction (0200) message with a processing code of "22 XX XX". Online voids of an online payment or offline refund are sent by a Financial Transaction (0200) message with a processing code of "23 XX XX". Online voids of an online cash advance or offline refund are sent by a Financial Transaction (0200) message with a processing code of "03 XX XX".

3.2.4 Referral Processing

If a voice authorization is required to authorize an Authorization Request (0100) message or a Financial Transaction Request (0200) message, the terminal will disconnect from the capture host.

When the Issuer authorizes the transaction after the terminal has disconnected from the capture host, the Issuer will provide an approval code verbally. This approval code is entered into the terminal by the Merchant and the terminal prepares a Financial Transaction Advice Request (0220) message containing the approval code.

However, the Financial Transaction Advice Request (0220) message may not immediately be transmitted to the host for processing, but is retained in the terminal.



3.2.5 Normal Advice Processing

When the Merchant initiates an amount adjustment (tip) or an offline transaction, the terminal will construct a Financial Transaction Advice Request (0220) message. However, this message is not immediately transmitted to the host for processing.

Offline transactions (advices) are stored until a successful Authorization Response (0110) or Financial Transaction Response (0210) message is received or a Reconciliation Request (0500) message is sent.

Personal Identifiable Information (PII) data is not present in advices. PII data is classified as; DE 35 Track2, DE 45 Track 1 Data, CVV2/CID Data.

3.2.6 EMV Advice Processing (Only when requested by Acquirer)

Some acquirers / processors need to confirm that the EMV transaction response has been processed by the POS and the Chip Card, and have feedback about how the Chip card processed the Online Response.

When this confirmation is requested, the Merchant terminal has to send an EMV processing Advice after the online EMV transaction

The Merchant terminal will construct a Financial Transaction Advice Request (0220) message with the data of the original request and some data of the online response, then transmitted as any other advice.

The EMV Transaction final response is given by Terminal /Chip Card and according to it the value of the DE 38 (Authorization Identification Response) must be changed on the 0220 message according the following table.

For Online declined transactions, there is no need for an EMV Advice.

Data Field 38 Authorization Identification Response for EMV Advices

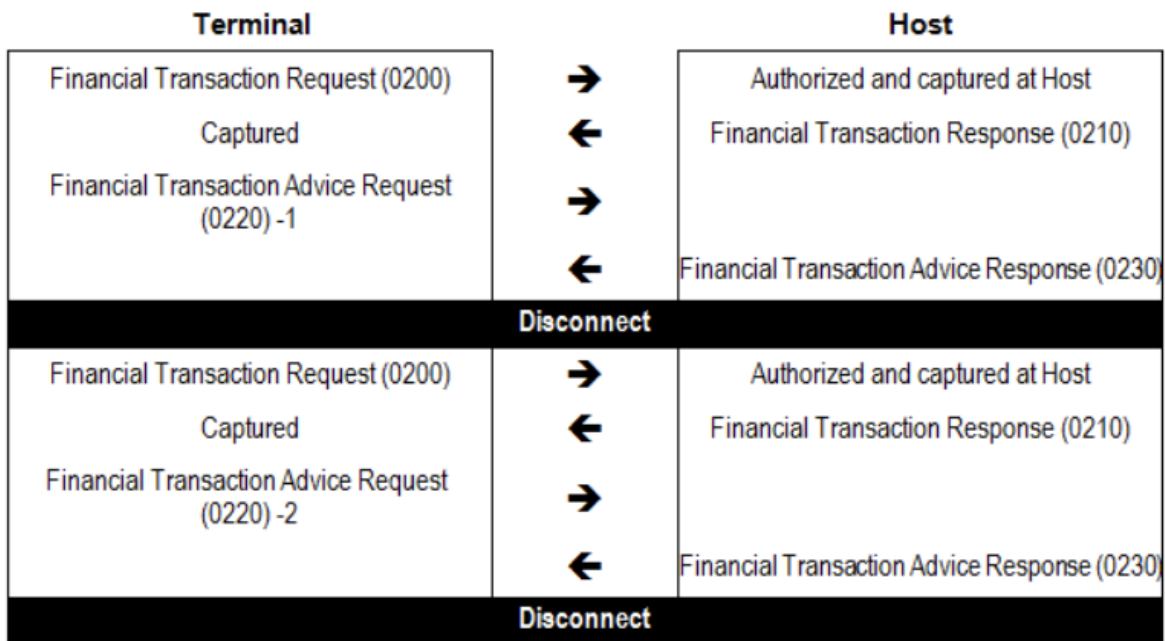
1st Cryptogram Type	Online Condition	2nd Cryptogram Type	DE 38 Value
TC	---	---	Y1
AAC	---	---	Z1
ARQC	Unable to go Online	TC	Y3
ARQC	Unable to go Online	AAC	Z3
ARQC	Approved Online	TC	Original Value found in response
ARQC	Approved Online	AAC	Z3

3.2.7 Trickle Feed of Advices

On receipt of a successful Authorization Response (0110) or Financial Transaction Response (0210) message, a single advice message is transmitted to the host during the same call. A Financial Transaction Advice Response (0230) message will be returned to the terminal.

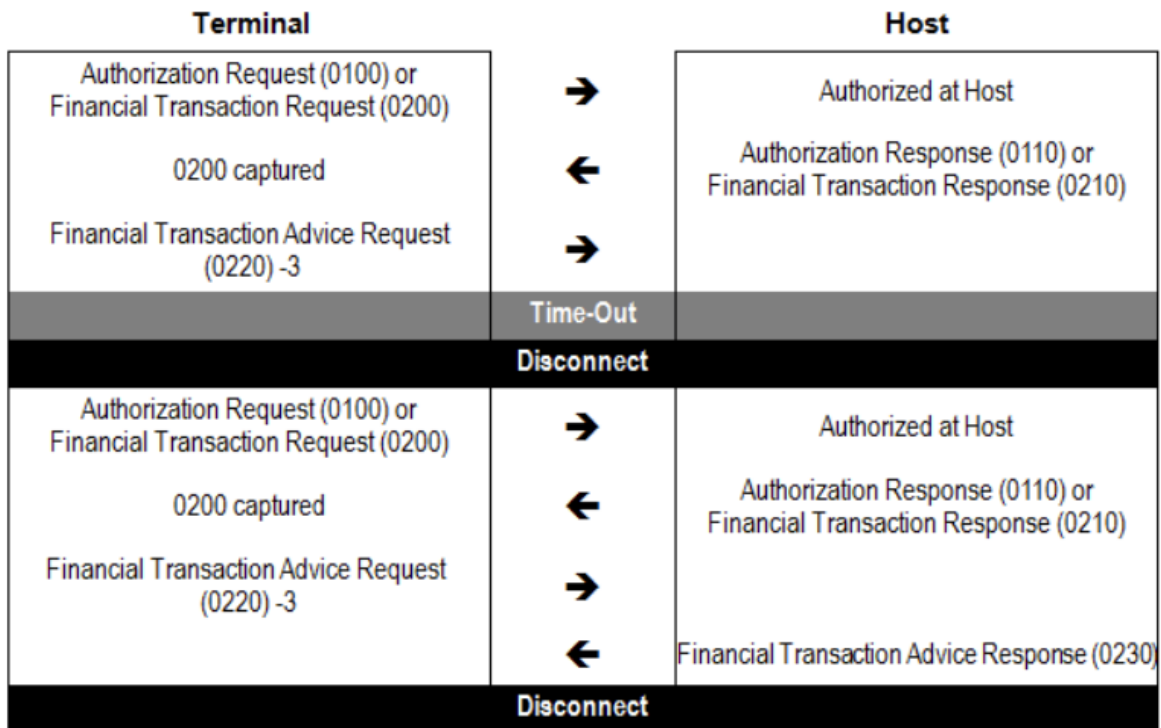
Offline Voids

If the terminal attribute allows for an offline void, they can be generated for offline sales, offline refunds, online sales and online refunds. Offline void is a Financial Advice Transaction (0220) message with the same processing code as an Online Void.



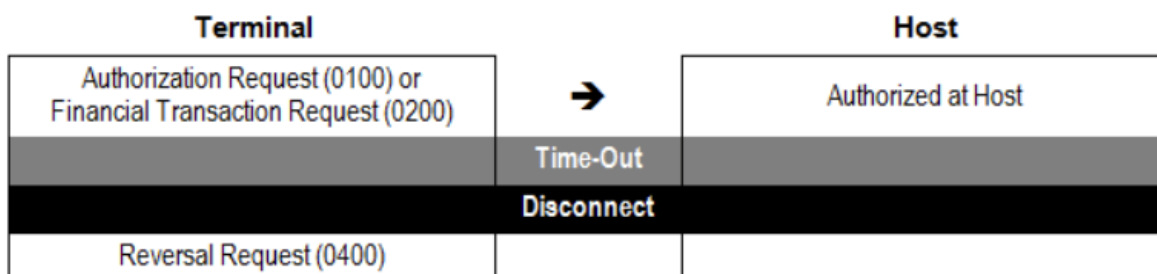
3.2.8 Advice Processing- Time-Out

If a time-out occurs during the transmission of a Financial Transaction Advice Request (0220) message, the advice will be sent again either after the next online transaction, such as a sale or authorization request or at the next attempt at submission.



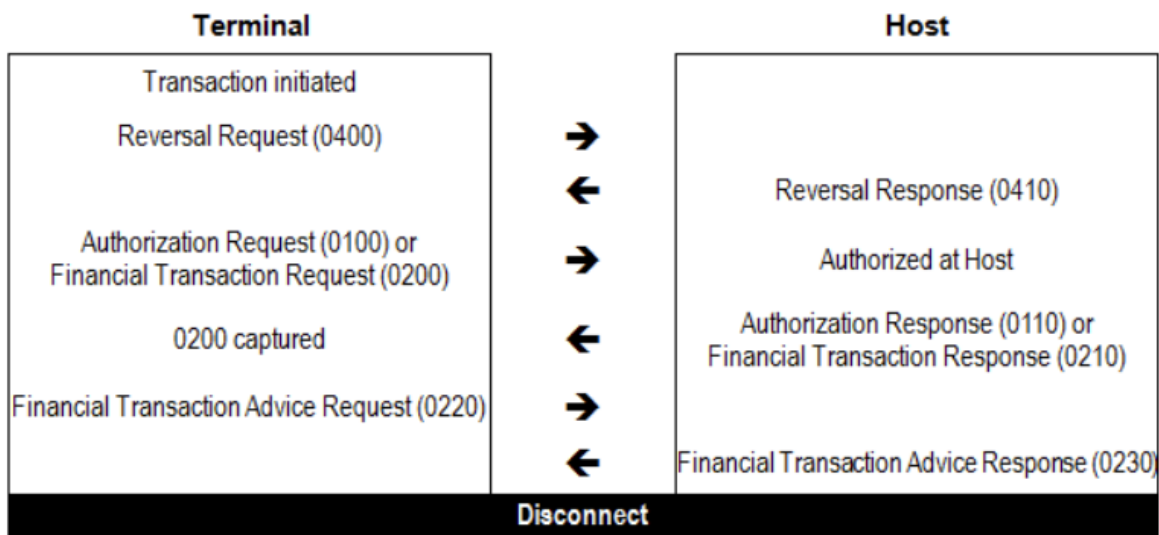
3.2.9 Reversal Processing

When a time-out occurs during the processing of an Authorization Request (0100) message or a Financial Transaction Request (0200) message, the host may or may not have received the message. Because the Merchant may re-submit the transaction, which could result in the creation of a duplicate, the terminal will create a Reversal Request (0400) message which must be sent to the host before processing another transaction.



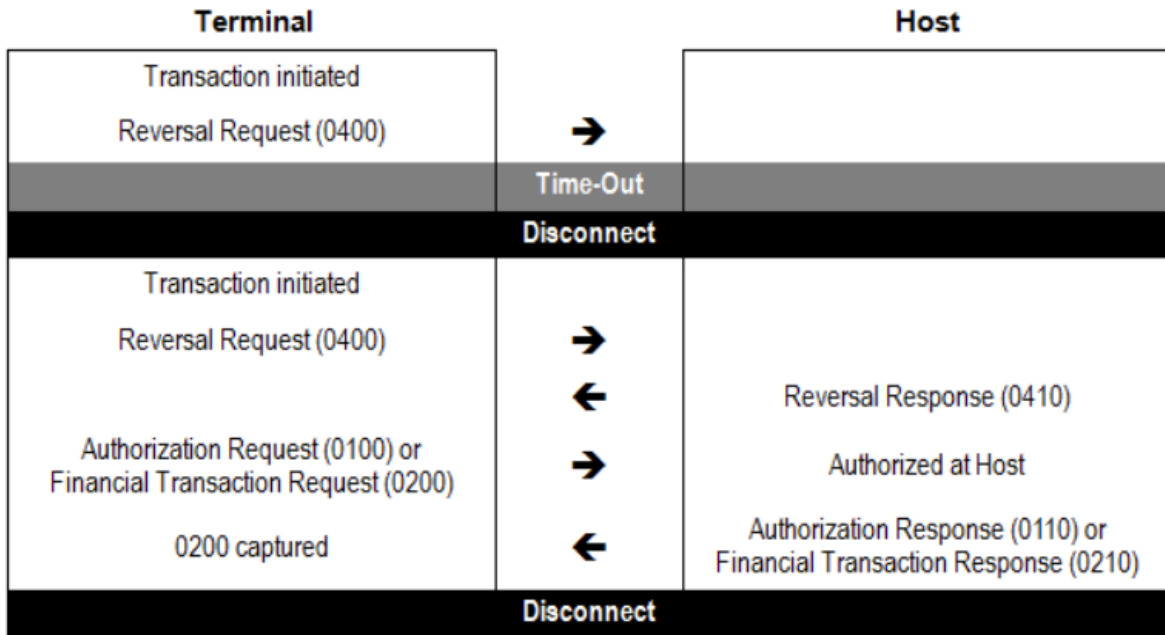
This reversal will cancel the original transaction at the host, if the host captured that transaction. The host will respond with a Reversal Response (0410) message with a 'reversal accepted' response code. If the time-out occurred before the host received the original transaction, the host will respond to the reversal request with a 'no action taken' response code.

Once the Reversal Response (0410) message is received at the terminal, the terminal is ready to process additional transactions



3.2.10 Reversal Processing - Time-Out

If a time-out occurs during the transmission of a reversal, no other online transactions will be transmitted to that host until a Reversal Response (0410) message is received. If a transaction is initiated at the terminal, the reversal will be re-sent prior to the transaction. The transaction will not be sent to the host until the Reversal Response (0410) message is received.



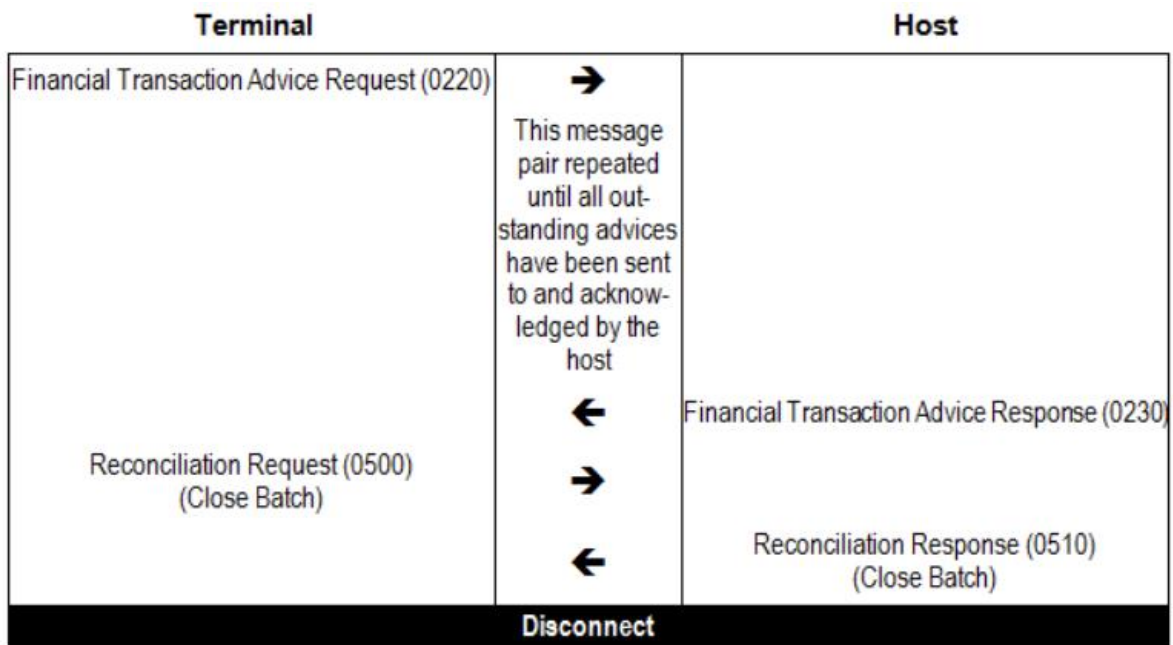
3.3 Reconciliation

Reconciliation messages are used to close a batch. The host maintains a copy (shadow file) of the terminal's batch, so that, at reconciliation, the terminal and the host compare totals. If the terminal and host totals match, closing a batch only requires the exchange of reconciliation messages (05xx). If the totals do not match, the reconciliation messages are used in conjunction with the batch upload messages (03xx).

3.3.1 Close Batch - Normal Processing

When a Merchant settles batch transactions at the terminal, a Reconciliation Request (0500) message is sent to the host containing submission details. Prior to sending the Reconciliation Request (0500) message, the terminal's journal is checked to ensure that all outstanding Financial Transaction Advice Request (0220) messages have been sent to and acknowledged by the host.

If the total number of transactions, total debits and total credits in the batch all balance between the terminal and the host, the host sends a Reconciliation Response (0510) message with an 'approved' response code and a text message for display to the Merchant.

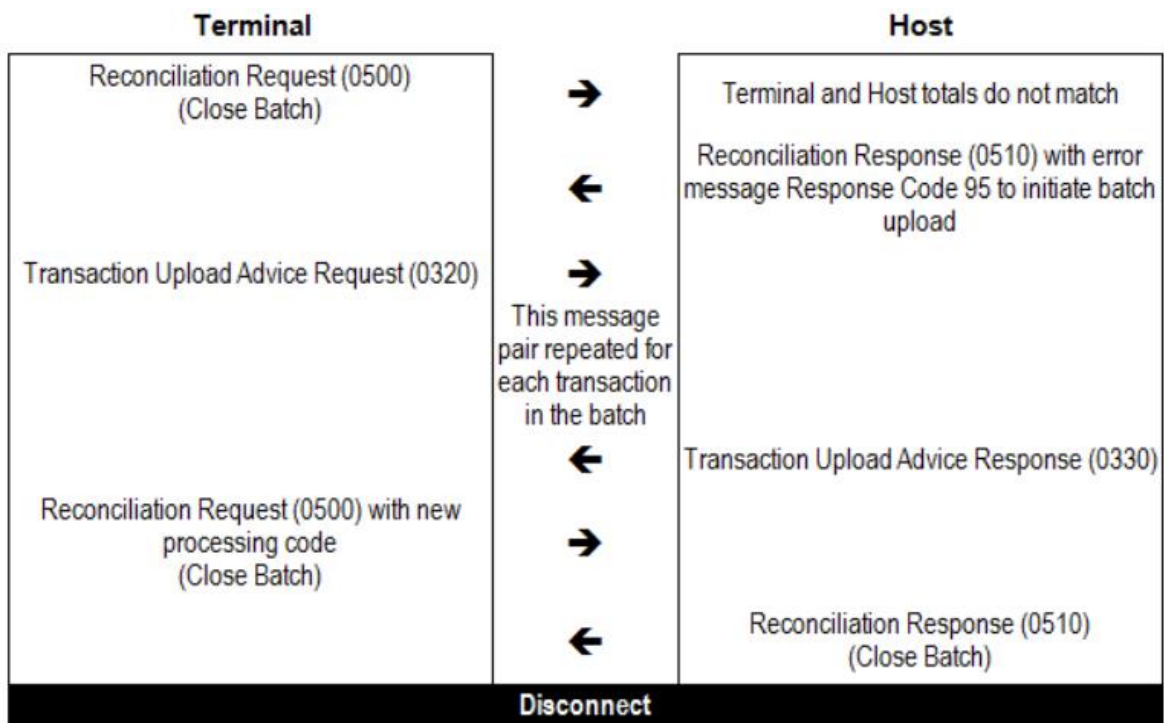


3.3.2 Close Batch - Error Processing

When a Reconciliation Request (0500) message is sent and the terminal totals do not match with the host, the host sends a response code “95” (Bad reconciliation - Send detail). The terminal then sends a Transaction Upload Advice Request (0320) message to the host, containing details for the first transaction in the batch and the host responds with a Transaction Upload Advice Response (0330) message containing a processing code that prompts the terminal to send the next transaction, if another exists. This 0320/0330 exchange continues until the whole batch has been uploaded.

When the host has received all of the transactions in the batch, reconciliation can start again. Because the host's shadow file will now contain the same transactions as the terminal, the reconciliation should be successful and the batch should be closed.

If the Reconciliation Response (0510) message contains a response code other than “00” (Approved/Accepted) or “95” (Bad reconciliation - Send detail), the appropriate error message will be displayed. Any subsequent attempt at a transaction for the host will be prohibited until a successful batch close occurs. If attempted, the terminal will sound its alarm and will display “RETRY BATCH” message.

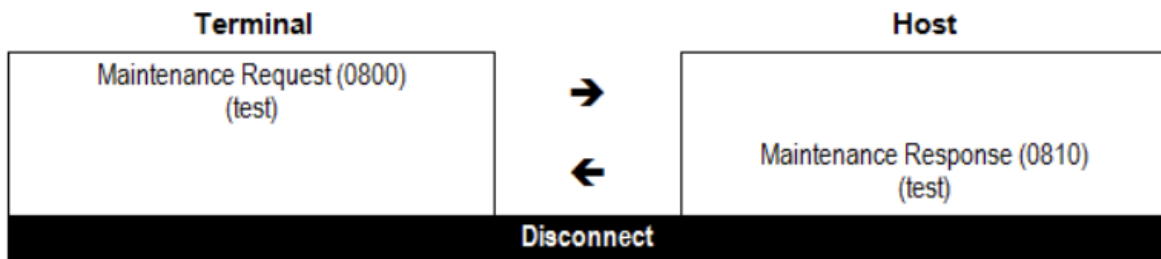


3.3.3 Close Batch - Time-Out

If a time-out occurs at any point during reconciliation, the terminal will display the appropriate message and disconnect the line. Any subsequent attempt at a transaction for the host will be prohibited until a successful batch close occurs. The Merchant will recommence the process and the terminal will restart Close Batch - Normal Processing.

3.4 Test Transaction

The Merchant may initiate a communications test with the “ISSwitch Server” host by selecting the relevant terminal function. The terminal will send a Maintenance Request (0800) message. If available, the host will respond with a Maintenance Response (0810) message.



4.0 Electronic Data Capture (EDC) Interface Messages

4.1 ISO 8583 Message Types

The following table lists all ISO 8583 Message Types used by POSARE Server.

MTI *	ISO 8583 Message Type	Message Function / Purpose / Usage
0100	Authorization Request	Authorization, Pre-Authorization, Authorization Voids
0110	Authorization Response	Authorization, Pre-Authorization, Authorization Voids Responses
0200	Financial Transaction Request	Financial Transaction Request for Sales, Refunds & Voids
0210	Financial Transaction Response	Financial Transaction Response for Sales, Refunds & Voids
0220	Financial Transaction Advice Request	Financial Advice Request for Sale Completion, Offline Sales, Debit Adjustment, Credit Adjustment, Offline Refunds and Offline Voids.
0230	Financial Transaction Advice Response	Financial Advice Response for Sale Completion, Offline Sales, Debit Adjustment, Credit Adjustment, Offline Refunds and Offline Voids.
0320	Transaction Download Advice Request	Batch Upload Advice Request Terminal to Host for Debits & Credits
0330	Transaction Download Advice Response	Batch Upload Advice Response Terminal to Host for Debits & Credits
0400	Reversal Request	Reversal Request Online for Authorization, Pre-Authorization & Sales
0410	Reversal Response	Reversal Response Online for Authorization, Pre-Authorization & Sales
0500	Reconciliation Request	Reconciliation Request for Close Batch
0510	Reconciliation Response	Reconciliation Response for Close Batch
0800	Maintenance Request	Maintenance Message Request for Test Transaction, Batch Statistics
0810	Maintenance Response	Maintenance Message Response for Test Transaction, Batch Statistics

*MTI = Message Type Identifier

4.1.1 Notes for Message Data Specification

- For Data Type, **A** = Alpha, **N** = Numeric, **S** = Special Characters, **B** = Binary
- For Data Field Format, **BCD** = Binary Coded Decimal, **HEX** = Hexadecimal
- For Message Type Identifier (e.g., 0100, 0110, 0200, 0210, etc.), **M** = Mandatory, **O** = Optional, **C** = Conditional, **-** = N/A

4.1.2 0100 — Authorization Request Table

Purpose: The Authorization Request (0100) message is sent from a terminal to the Host for the following transactions:

- Card Authorization Request — An authorization is defined as an approval of a transaction given by the card Issuer. The terminal does not capture approved authorizations, only transactions for submission.
- Pre-Authorization Request — The pre-authorization transaction is also an approval of a transaction, but implies that additional charges may be authorized later for the same transaction, but prior to submitting the transaction.

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0100	RESPONSE 0110
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	C	O
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	M	C
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	—	M
13	Date, Local Transaction	4 bytes, fixed	N	BCD	—	M
14	Date, Expiration	2 bytes, fixed	N	BCD	C	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	M	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	M	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	C	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	—	M
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	—	C
39	Response Code	2 bytes, fixed	AN	HEX	—	M

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0100	RESPONSE 0110
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
42	Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
43	Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
44	Additional Response - Data	LLVAR	ANS	HEX	C	C
45	Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	C	—
48	Additional Data- Private	LLLVAR	ANS	HEX	—	—
49	Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
52	PIN Data	8 bytes	B	HEX	O	—
54	Additional Amounts	12 bytes, max., LLLVAR	N	HEX	O	C
55	Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	C	C
60	Additional Data- Private	LLLVAR	ANS	HEX	—	—
61	Additional Data- Private	LLLVAR	ANS	HEX	O	—
62	Additional Data- Private	LLLVAR	ANS	HEX	M	—
63	Additional Data- Private	LLLVAR	ANS	HEX	O	C

4.1.3 0200 — Financial Transaction Request Table

Purpose: The Financial Transaction Request (0200) message is sent from a terminal to the Host for the following transactions:

- Sale
- Sale Void (Online)
- Cash Advance
- Cash Advance Void (Online)
- Refund
- Refund Void (Online)

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0200	RESPONSE 0210
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	C	O
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	M	C
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	—	M
13	Date, Local Transaction	4 bytes, fixed	N	BCD	—	M
14	Date, Expiration	2 bytes, fixed	N	BCD	C	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	M	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	M	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	C	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	C**	M
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	C**	C
39	Response Code	2 bytes, fixed	AN	HEX	—	M
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
42	Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
43	Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
44	Additional Response - Data	LLVAR	ANS	HEX	C	C
45	Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	C	—
48	Additional Data- Private	LLLVAR	ANS	HEX	—	—
49	Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
52	PIN Data	8 bytes	B	HEX	O	—
54	Additional Amounts	12 bytes, max., LLLVAR	N	HEX	O	C

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0200	RESPONSE 0210
55	Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	C	C
60	Additional Data- Private	LLLVAR	ANS	HEX	—	—
61	Additional Data- Private	LLLVAR	ANS	HEX	O	—
62	Additional Data- Private	LLLVAR	ANS	HEX	M	—
63	Additional Data- Private	LLLVAR	ANS	HEX	O	C

** Conditional Fields 37 and 38 are required for Void Transactions. The data received in the 0210/0110 response from the host must be used to fill these fields in the Void Request.

4.1.4 0220 — Financial Transaction Advice Request Table

Purpose: The Financial Transaction Advice Request (0220) message is sent from a terminal to the Host for the following transactions:

- Sale Completion
- Offline Sale
- Debit Adjustment
- Credit Adjustment
- Offline Refund
- Offline Void Refund

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0220	RESPONSE 0230
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	M	O
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	M	C
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	M	—
13	Date, Local Transaction	4 bytes, fixed	N	BCD	M	—
14	Date, Expiration	2 bytes, fixed	N	BCD	M	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	M	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	M	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	—	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	C **	M
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	M **	—
39	Response Code	2 bytes, fixed	AN	HEX	—	M
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
42	Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
43	Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
44	Additional Response - Data	LLVAR	ANS	HEX	—	C
45	Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	—	—
48	Additional Data- Private	LLLVAR	ANS	HEX	—	—
49	Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
54	Additional Amounts	12 bytes, max., LLLVAR	N	HEX	O	—

55	Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	—	—
Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0220	RESPONSE 0230
60	Additional Data- Private	LLLVAR	ANS	HEX	M	—
61	Additional Data- Private	LLLVAR	ANS	HEX	O	—
62	Additional Data- Private	LLLVAR	ANS	HEX	M	—
63	Additional Data- Private	LLLVAR	ANS	HEX	O	C

** Fields 37 and 38 are required for Adjust/Sale Completion Advice Transactions. The data received in the 0210/0110 response from the host must be used to fill these fields in the Void Request.

For Offline Sale transactions only the Field 38 is required.

4.1.5 0320 — Transaction Upload Advice Request Table

Purpose: The Transaction Upload Advice Request (0320) message is used to upload the transactions from a non-balancing batch in a POS device to the Capture Host.

Transaction Upload Advice messages are triggered with the receipt of a Reconciliation Response (0510) (original Reconciliation Request (0500) with the Processing Code (Data Field 3) of “92 xx xx”) with a Response Code (Data Field 39) of “95” (Bad reconciliation - Send detail).

The POS device will then send Transaction Upload Advice (0320) requests to the Capture Host for each transaction from the batch being processed.

The Transaction Upload Advice Response (0330) acknowledges that the Capture Host received the transaction. Once all the transactions in the batch are uploaded, the POS device will then submit a Batch Close After Upload transaction (Reconciliation Request (0500) message with the Processing Code (Data Field 3) of “96 xx xx”).

Note: If a transaction in the batch has been adjusted or voided, the upload transaction is a single composite of the original message data with the resulting changes of the adjustment or void

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0320	RESPONSE 0330
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	M	O
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	M	C
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	M	M
13	Date, Local Transaction	4 bytes, fixed	N	BCD	M	M
14	Date, Expiration	2 bytes, fixed	N	BCD	M	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	M	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	M	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	—	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	C	M
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	M	O

39	Response Code	2 bytes, fixed	AN	HEX	—	M
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
Data Field #				Data Field Format	REQUEST 0320	RESPONSE 0330
	Data Field Name	Data Field Length	Data Type			
	42 Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
	43 Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
	44 Additional Response - Data	LLVAR	ANS	HEX	C	C
	45 Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	—	—
	48 Additional Data- Private	LLLVAR	ANS	HEX	—	—
	49 Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
	54 Additional Amounts	12 bytes, max., LLLVAR	N	HEX	O	—
	55 Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	—	—
	60 Additional Data- Private	LLLVAR	ANS	HEX	—	—
	61 Additional Data- Private	LLLVAR	ANS	HEX	O	—
	62 Additional Data- Private	LLLVAR	ANS	HEX	M	—
	63 Additional Data- Private	LLLVAR	ANS	HEX	O	C

4.1.6 0400 — Reversal Request Table

Purpose: The Reversal Request (0400) message should be created by the electronic medium used to enter the original Financial Transaction Request (0200) message or Authorization Transaction Request (0100) message.

Only the original data field values used to generate the original Transaction Request message can be used to populate the data field values in the Reversal Request (0400) message.

The Reversal Request (0400) message is an online message sent from a terminal to the Host to request reversal of previously submitted transactions.

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0400	RESPONSE 0410
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	M	M
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	M	O
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	—	M
Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0400	RESPONSE 0410
13	Date, Local Transaction	4 bytes, fixed	N	BCD	—	M
14	Date, Expiration	2 bytes, fixed	N	BCD	M	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	M	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	M	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	—	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	—	M
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	—	C
39	Response Code	2 bytes, fixed	AN	HEX	—	M
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
42	Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
43	Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
44	Additional Response - Data	LLVAR	ANS	HEX	C	C
45	Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	—	—

48	Additional Data- Private	LLLVAR	ANS	HEX	—	—
49	Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
54	Additional Amounts	12 bytes, max., LLLVAR	N	HEX	O	—
55	Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	—	—
60	Additional Data- Private	LLLVAR	ANS	HEX	—	—
61	Additional Data- Private	LLLVAR	ANS	HEX	O	—
62	Additional Data- Private	LLLVAR	ANS	HEX	M	—
63	Additional Data- Private	LLLVAR	ANS	HEX	O	—

4.1.7 0500 — Reconciliation Request Table

Purpose: The terminal uses reconciliation messages to settle transactions with the settlement host. When a terminal reconciles its transactions with the Capture Host, it performs a procedure called closing.

A batch is closed in the terminal, when all transactions in the batch are reconciled with the host. Once a batch is closed, the terminal cannot longer access it; and the terminal must delete its copy of the batch from the terminal's memory.

The following actions are acceptable for the Reconciliation Request (0500) message:

- Close Batch
- Close Batch Following Batch Upload

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0500	RESPONSE 0510
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	—	—
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	—	—
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	—	M
13	Date, Local Transaction	4 bytes, fixed	N	BCD	—	M
14	Date, Expiration	2 bytes, fixed	N	BCD	—	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	—	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	—	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	—	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	—	M
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	—	M
39	Response Code	2 bytes, fixed	AN	HEX	—	M
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
42	Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
43	Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
44	Additional Response - Data	LLVAR	ANS	HEX	—	—
45	Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	—	—
48	Additional Data- Private	LLLVAR	ANS	HEX	—	—

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0500	RESPONSE 0510
49	Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
54	Additional Amounts	12 bytes, max., LLLVAR	N	HEX	—	—
55	Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	—	—
60	Additional Data- Private	LLLVAR	ANS	HEX	M	—
61	Additional Data- Private	LLLVAR	ANS	HEX	—	—
62	Additional Data- Private	LLLVAR	ANS	HEX	—	—
63	Additional Data- Private	LLLVAR	ANS	HEX	M	M

4.1.8 0800 — Maintenance Request

Purpose: The Maintenance Request (0800) message can be sent for the following requests:

- Echo Test

Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0800	RESPONSE 0810
—	Message Type Identifier	2 bytes, fixed	N	BCD	M	M
—	Bit Map - Primary	8 bytes, 64 bits	B	HEX	M	M
2	Primary Account Number (PAN)	21 bytes, max., LLVAR	N	BCD	—	—
3	Processing Code	3 bytes, fixed	N	BCD	M	M
4	Amount, Transaction	6 bytes, fixed	N	BCD	—	—
7	Transmission Date and Time	5 bytes, fixed	N	BCD	—	—
11	Systems Trace Audit Number	3 bytes, fixed	N	BCD	M	M
12	Time, Local Transaction	3 bytes, fixed	N	BCD	—	O
13	Date, Local Transaction	4 bytes, fixed	N	BCD	—	O
14	Date, Expiration	2 bytes, fixed	N	BCD	—	—
18	Merchant Type	4 bytes, fixed	N	BCD	—	—
22	Point of Service Entry Mode	2 bytes, fixed	N	BCD	—	—
24	Network International Identifier (NII)	2 bytes, fixed	N	BCD	M	—
25	Point of Service Condition Code	1 byte, fixed	N	BCD	—	—
35	Track 2 Data	38 bytes, max., LLVAR	ANS	HEX	—	—
37	Retrieval Reference Number (RRN)	12 bytes, fixed	AN	HEX	—	O
38	Authorization Identification Response	6 bytes, fixed	AN	HEX	—	—
39	Response Code	2 bytes, fixed	AN	HEX	—	M
41	Card Acceptor Terminal Identification	8 bytes, fixed	ANS	HEX	M	M
Data Field #	Data Field Name	Data Field Length	Data Type	Data Field Format	REQUEST 0800	RESPONSE 0810
42	Card Acceptor Identification Code	15 bytes, fixed	ANS	HEX	M	O
43	Card Acceptor Name/Location	40 bytes, fixed	ANS	HEX	—	—
44	Additional Response - Data	LLVAR	ANS	HEX	—	—
45	Track 1 Data	77 bytes, max., LLVAR	ANS	HEX	—	—
48	Additional Data- Private	LLLVAR	ANS	HEX	—	—
49	Currency Code, Transaction	3 bytes, fixed	N	BCD	—	—
54	Additional Amounts	12 bytes, max., LLLVAR	N	HEX	—	—
55	Integrated Circuit Card System Related Data - EMV	LLLVAR	ANS	HEX	—	—

60	Additional Data- Private	LLVAR	ANS	HEX	—	—
61	Additional Data- Private	LLVAR	ANS	HEX	—	—
62	Additional Data- Private	LLVAR	ANS	HEX	—	—
63	Additional Data- Private	LLVAR	ANS	HEX	—	—

5.0 Overview of ISO 8583 Messages

The *ISO 8583* standard defines a bit-mapped message format. The first fifteen bytes of a message are composed of TPDU header, a four-byte message type code that identifies the type of transaction being transmitted and an eight-byte bit map that indicates the data fields that immediately follow. These two data fields always precede the transaction data in each message.

The bit map (which indicates the data fields used in a message) is eight bytes (64 bits) long. Each bit represents a data field defined in this specification, and contains either the value “1” to indicate the presence of the data field or a “0” (zero) to indicate its absence. The binary indicators are then translated to hexadecimal notation.

This specification contains information on each transaction used by “ISSwitch Server” EDC interface, and only those data fields used are included. Additional information on the ISO standard and/or ISO data field definitions is available in International Standard ISO 8583.

Also, some Private Use data fields may be redefined for different transactions, depending on the message type specified. For example, Private Use Data, Data Field 63 may contain batch count and amount subfields for some message types and industry-specific transaction information subfields for others.

5.1 ISO 8583 General Message Structure

ISO 8583 format messages are constructed as specified in the ISO 8583-1987 standard.

ISO 8583 standard provides for variable length messages that are bit map driven.

A bit map consists of a 64-bit string contained within an eight-byte data field.

The data content of a message is determined by the value (1) or (0) of bits in a bit map data field. Each bit is associated with a unique data field.

Element	TPDU			ISO 8583 Message Data		
	ID	Destination Address	Originator Address	MTI	Bitmap	Data Elements
Length	1	2	2	2	8 .. 16	var ...
	5			var ...		

5.1.1 Transport Protocol Data Unit (TPDU)

The TPDU contains addressing information related to both the transaction destination (host application process or network address) and the transaction-originating device (Terminal or POS system). The TPDU is a 5-byte header that precedes the application data.

5.1.1.1 Request TPDU

TPDU Id	Identifies TPDU type 60h - Transactions 68h - NMS/TNMS
Destination Address	Network International Identifier
Originator Address	Identifies the individual terminal or process originating the transaction.

5.1.1.2 Response TPDU

TPDU Id	Identifies TPDU type Same value as in the request message.
Destination Address	Same as Originator address from request message.
Originator Address	Same as Destination address in request message.

5.1.2 Message Type Identifier (MTI)

Each digit of the message type or MTI code identifies a message attribute.

Definitions of the attributes are:

- First position / Version Number
This number is always “0” (zero).
- Second position / Message Class
 - 1 = Authorization
 - 2 = Financial Capture
 - 3 = File Update
 - 4 = Reversals
 - 5 = Reconciliation
 - 8 = Maintenance
- Third position / Message Function
 - 0 = Request
 - 1 = Response to a Request
 - 2 = Advice
 - 3 = Response to an Advice
 - 4 = Notification
 - 5-9 = Reserved for ISO use
- Fourth position / Transaction Originator
This digit is always "0" (zero) for POS originated transactions

MTI	ISO 8583 Message Type	Message Function / Purpose / Usage
0100	Authorization Request	Authorization, Pre-Authorization, Authorization Voids, Travelers Cheque and Emergency Check Cashing Requests
0110	Authorization Response	Authorization, Pre-Authorization, Authorization Voids & Travelers Cheque Responses
0200	Financial Transaction Request	Financial Transaction Request for Sales, Refunds & Voids
0210	Financial Transaction Response	Financial Transaction Response for Sales, Refunds & Voids
0220	Financial Transaction Advice Request	Financial Advice Request for Sale Completion, Offline Sales, Debit Adjustment, Credit Adjustment, Offline Refunds and Offline Voids.
0230	Financial Transaction Advice Response	Financial Advice Response for Sale Completion, Offline Sales, Debit Adjustment, Credit Adjustment, Offline Refunds and Offline Voids.
0320	Transaction Download Advice Request	File Update Advice Request Terminal to Host for Debits & Credits
0330	Transaction Download Advice Response	File Update Advice Response Terminal to Host for Debits & Credits

0400	Reversal Request	Reversal Request Online for Authorization, Pre-Authorization & Sales
0410	Reversal Response	Reversal Response Online for Authorization, Pre-Authorization & Sales
0500	Reconciliation Request	Reconciliation Request for Close Batch
0510	Reconciliation Response	Reconciliation Response for Close Batch
0800	Maintenance Request	Maintenance Message Request for Test Transaction, Initialize Terminal & Batch Statistics
0810	Maintenance Response	Maintenance Message Response for Test Transaction, Initialize Terminal & Batch Statistics

5.1.3 ISO 8583 Message Bit Map

The third component of a message (after the Message Type Identifier (e.g., 0200) is a bit map). The bit map consists of 64 bits, numbered from left to right starting with bit 1. Each bit in the bit map signifies the presence (1) or the absence (0) in the message of its corresponding data element (e.g., if bit 35 contains a “1”, the Track 2 Data field will be present in the message).

The Primary Bit Map identifying the presence or absence of the data fields represented by bits 1-64, will always be present in a message. Bit 1 in the Primary Bit Map will indicate the presence or absence of a Secondary Bit Map (bits 65-128).

5.2 Guidelines for Using the ISO 8583 Message Formats

Unless otherwise specified, all fixed-length numeric data fields should be right justified and zero filled, fixed-length alphanumeric data fields should be left justified and character space filled, and binary data fields should be in eight-bit blocks that are left justified and zero filled.

Data field requirements are as follows:

Mandatory (M): Data field and contents are required to process this message. Data field must contain the appropriate text or numeric information as indicated.

Optional (O): Data field and contents are not mandatory for processing the message, but should be provided if available.

Conditional (C): A data field may be conditional if it is only used in certain circumstances. See Data Field Descriptions for specific details. For example, Data Fields 2 and 14 are conditional if the Merchant's system is unable to provide track data intact.

Bit	This column shows the relative position in the bit map, of the bit representing the data element
Name	The name of the data element.
Format	The format of data elements shown in the primary bit map are represented as follows: L = Length of variable data field (less than 10 characters in length) that follows. LL = Length of variable data field (less than 100 characters in length) that follows. □ This data field will be packed and occupy 1 byte. LLL = Length of variable data field (less than 1000 characters in length) that follows. This data field will be packed and occupy 2 bytes. VAR = Variable length data field.
Attribute	The abbreviations used to describe the data element are as follows: A = Alphabetic characters. AN = Alphabetic and numeric characters. ANS = Alphabetic, numeric and special characters. B = Binary representation. BCD = Binary coded decimal. N = Numeric digits.

5.4 Data Field Descriptions

Data Field — MTI

MESSAGE TYPE IDENTIFIER

Length of Field: 2 bytes, fixed
Field Type: Numeric
Field Format: Binary coded decimal (BCD)

Field Requirement:

MTI	
0100	M
0110	M
0200	M
0210	M
0220	M
0230	M
0320	M
0330	M
0400	M
0410	M
0500	M
0510	M
0800	M
0810	M

Description:

This data field contains a Message Type Identifier (MTI), which is a four-digit code that indicates the type of data contained in this message.

Valid entries include the following:

0100= Authorization Request
0110= Authorization Response
0200= Financial Transaction Request
0210= Financial Transaction Response
0220= Financial Transaction Advice Request
0230= Financial Transaction Advice Response
0320= Transaction Upload Advice Request
0330= Transaction Upload Advice Response
0400= Reversal Request
0410= Reversal Response
0500= Reconciliation Request
0510= Reconciliation Response
0800= Maintenance Request
0810= Maintenance Response

Data Field Bitmap**BITMAP PRIMARY**

Length of Field: 8 bytes, 64 bits, fixed length for each bit map
Field Type: Binary
Field Format: Hexadecimal
Field Requirement:

MTI	
0100	M
0110	M
0200	M
0210	M
0220	M
0230	M
0320	M
0330	M
0400	M
0410	M
0500	M
0510	M
0800	M
0810	M

Description:

The bit map consists of 64-bits, numbered from left to right starting with bit 1.

Each bit in this data element signifies the presence (value 1) or absence (value 0) of the respective data field.

If a data field is present, its assigned bit map position must contain a value of "1". If the data field is not present, its assigned bit map position must contain a value of "0".

Data Field 2**PRIMARY ACCOUNT NUMBER (PAN)**

Length of Field: 3 bytes minimum, 21 bytes maximum
(LLVAR)

Variable Len.

Indicator: 2 bytes, BCD

Field Type: Numeric

Field Format: Binary Coded Decimal (BCD)

Field Requirement:

010 0	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	----	C	----	M	----	M	----	M	----	----	----	----	----

Description:

This data field contains the Primary Account Number (a.k.a., PAN or Card Number), preceded by a two-digit, Variable Length Indicator (VLI).

The VLI must indicate the exact length of the account number.

The VLI and the PAN data are both transmitted in BCD format.

If the account number is by example 13 bytes (an odd length), which must be padded with a trailing "F" or "0"; and 14 digits of variable data are actually transmitted.

If the account number is by example 16 bytes (an even length), no padding is needed; and 16 digits of variable data are actually transmitted.

VLI	PAN	Data to send
13	4123456789012	1341234567890120
16	4123456789012340	164123456789012345

Encoded BCD Bytes									
1	2	3	4	5	6	7	8	9	BYTE
13	41	23	45	67	89	01	20		HEX CONTENT
16	41	23	45	67	89	01	23	45	HEX CONTENT

Data Field 3**PROCESSING CODE**

Length of Field: 3 bytes, fixed
Field Type: Numeric
Field Format: Binary Coded Decimal (BCD)

Field Requirement:

MTI	
0100	M
0110	M
0200	M
0210	M
0220	M
0230	M
0320	M
0330	M
0400	M
0410	M
0500	M
0510	M
0800	M
0810	M

Description: This data field contains the Processing Code, indicating the financial service being requested. Codes are derived from ISO 8583; however, the last digit in this entry is used as a flow control indicator when multiple records are transmitted. Valid processing and flow control codes include those listed in the table that follows:

Data Field 3**PROCESSING CODE (Continued)**

MTI	Processing Code	Transaction
0100	00 aa 0x	Authorization
	31 aa 0x	Balance Inquiry
	38 aa 0x	Card Verification
	81 aa 0x	Loyalty Balance Inquiry (Def 1.0)
0200	00 aa 0x	Sale
	01 aa 0x	Cash Advance
	09 aa 0x	Sale + Cash
	20 aa 0x	Refund
	21 aa 0x	Payment
		Loyalty Redemption (Points / Miles)
	81 aa 0x	
	02 aa 0x	Sale VOID
	03 aa 0x	Cash Advance VOID
	11 aa 0x	Sale + Cash VOID
	22 aa 0x	Refund Void
	23 aa 0x	Payment Void
	83 aa 0x	Loyalty Redemption Void
0220	00 aa 0x	Offline Sale
	02 aa 0x	Adjust
	20 aa 0x	Offline Refund
0320	Same as Original	Batch Upload
0400	Same as Original	Reversal
0500	92 00 0x	Settlement Request
	96 00 0x	Settlement Close, After Upload
0800	92 00 0x	Logon
	99 00 0x	Test Transaction

*** Value specified by “aa” in the processing code table corresponds to an **Account Code** in the following table.

Data Field 3 PROCESSING CODE - Account Selection

Account Code	Account Selected
00	Default Account
10	Savings Account
20	Checking Account
30	Credit Card
40	Universal Account
Non Standard Account	
05	Loyalty Account for VASA Redemption
95	Loyalty Account for Financial Exchange

**Data Field 3 PROCESSING CODE - Flow Control Digit X
Substitute de "X" in the preceding tables**

Value	Action
0	No more messages to send
1	More messages to follow, hold the line
2	Parameter Update Pending, RFU
3	Prepare for batch upload

Data Field 4**AMOUNT, TRANSACTION**

Length of Field: 6 bytes, fixed
 Field Type: Numeric, right justified, zero filled
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	C	M	-----	M	-----	M	-----	M	-----	-----	-----	-----	-----

Description: This data field contains the total Transaction Amount (including taxes and tip).
 Two decimal places are implied. Thus, the value 100.00 would be entered as:
 "000000010000"
 For Void transactions, Amount must be "zero"
 (00 00 00 00 00 00).

Data Field 7**TRANSMISSION DATE AND TIME**

Length of Field: 5 bytes, fixed
 Numeric,
 Field Type: MMDDhhmmss
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
O	O	O	O	O	O	O	O	O	O	O	O	O	O

Description: This data field contains the Transmission Date and Time, which is the system date and time (e.g., GMT-6) when the Merchant transmits the transaction information to the Host. The format is MMDDhhmmss. This entry must be a valid date and time.

Data Field 11**SYSTEMS TRACE AUDIT NUMBER**

Length of Field: 3 bytes, fixed
 Field Type: Numeric
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

010 0	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
M	M	M	M	M	M	M	M	M	M	M	M	M	M

Description: This data field contains the Systems Trace Audit Number (STAN), which is a POS device/Merchant system-generated sequential number that uniquely identifies a transaction. This sequence number should be incremented for each request message transmitted to the "ISSwitch Server" Host; and when "99 99 99" is reached, the following request should restart at "00 00 01".
 * Zero ("00 00 00") is an invalid value and must not be used.
 ** For Reversal Request (0400) message only, the System Trace Audit Number entered in this data field must be the same value used in the original 0100, 0200 request that is being reversed.

Data Field 12**TIME, LOCAL TRANSACTION**

Length of Field: 2 bytes, fixed
 Field Type: hhmmss
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
O	M	O	M	M	M	M	C	O	O	O	O	O	O

Description: This data field contains the original Transaction Time, which is the local time when the transaction took place at the card acceptor location. The format is hh mm ss, and the value must be a valid time.
 Note: This original value should be used in subsequent transactions (i.e., adjustments, etc.).

Data Field 14**DATE, EXPIRATION**

Length of Field: 2 bytes, fixed
 Numeric, YY

Field Type: MM

Field Format: Binary Coded Decimal (BCD)

Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	---	C	---	M	---	M	---	M	---	---	---	---	---

Description: This data field contains the Expiration Date embossed on the face of the Card.

Note that most Card products are embossed with dates in format MM YY, which may require that the data entered in this data field be converted by reversing the month and year values from the Card, so that this entry appears in format YY MM.

*This data field is not required if the message contains either Track 1 , Track 2 data read from the Card's magnetic stripe, or ICC data.

Data Field 15**DATE, SETTLEMENT**

Length of Field: 2 bytes, fixed
 Numeric, MM

Field Type: DD

Field Format: Binary Coded Decimal (BCD)

Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
---	---	---	---	---	---	---	---	---	---	O	O	---	---

Description: This data field contains the Transaction Settlement Date. This optional data field may be used to transmit a Summary of Charge (SOC) batch settlement date.

If data is submitted, the format is MM DD, and the value must be a valid date.

Data Field 22**POINT OF SERVICE ENTRY MODE**

Length of Field: 2 bytes, fixed
Field Type: Numeric
Field Format: Binary Coded Decimal (BCD)
Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
M	---	M	---	M	---	M	---	M	---	---	---	---	---

Description:

This data field contains the Point of Service (POS) Entry Mode code , which is a series of values that identify terminal capability, security data and specific conditions present at the time the transaction occurred at the point of service.

This three-byte code is transmitted in BCD format, and the odd length must be padded. For example, “022”, must be padded with a leading zero to create the even-length value “00 22”, and four digits of data are actually transmitted.

The POS Entry Mode code is composed of two sub codes. Positions 1 and 2 indicate the Card Data Input Capability of the POS device (e.g., Magnetic Stripe Read, Integrated Circuit Card (ICC), etc.).

Similarly, Position 3 indicates the Cardholder Authentication Capability (i.e., the primary means used to verify the Card member’s identity (e.g., PIN entry).

POS devices should always attempt to populate all positions in the POS Entry Mode code with valid data. However, if the applicable information is unavailable or unknown, the Merchant should consult with their terminal provider to determine the appropriate value(s) to use.

Data Field 22	POS Entry Mode - Position 1-2 - Terminal Capability
00	Unspecified
01	Manual/Keyed Entry
02	Magnetic Stripe
05	ICC read
80	Magnetic stripe even though it is ICC capable a.k.a. Fallback
Amex Exclusive Values	
51	EMV Terminal - Manual Key Entry
52	EMV Terminal - Magnetic Stripe Entry
55	EMV Terminal - ICC read
56	EMV Terminal - Magnetic Stripe + 4DBC
57	EMV Terminal - Manual Key + 4DBC
62	EMV Terminal - Fallback + Magnetic Stripe
66	EMV Terminal - Fallback + Magnetic Stripe + 4DBC
Data Field 22	POS Entry Mode - Position 3 - Pin Entry Capability
0	Unspecified
1	PIN Entry Capability
2	NO PIN Entry Capability
Amex Exclusive Values	
3	PIN Entry Capability Offline

Data Field 24**NETWORK INTERNATIONAL IDENTIFIER (NII)**

Length of Field: 2 bytes, fixed
 Field Type: Numeric
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
M	O	M	O	M	O	M	O	M	O	M	O	M	O

Description: This data field contains the Network International Identifier (NII) code that identifies the processing host from the international network of card processors.
 This three-byte code is transmitted in BCD format, and the odd length must be padded. For example, "003" must be padded with a leading zero to create the even-length value "00 03", and four digits of data are actually transmitted.

Data Field 25**POINT OF SERVICE CONDITION CODE**

Length of Field: 1 bytes, fixed
 Field Type: Numeric
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
M	---	M	---	M	---	M	---	M	---	---	---	---	---

Description: This data field contains the POS (Point of Service) Condition Code, which indicates the condition under which the transaction took place at the point of sale/service. See the following valid codes.

Data Field 25 POS CONDITION CODE - Values

00	Normal presentment
05	Card not present
06	Pre-authorized request
08	Mail and/or telephone order

Data Field 35**TRACK 2 DATA**

Length of Field: 2 bytes minimum, 38 bytes maximum, (LLVAR)
 Variable Len.
 Indicator: 1 byte BCD, right justified, zero filled
 Field Type: Alphanumeric (upper case) & special characters
 Field Format: Binary Coded Decimal (BCD)
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	---	C	---	---	---	---	---	---	---	---	---	---	---

Description: This data field contains the information encoded in a valid Track 2 read, preceded by a two-digit, Variable Length Indicator (VLI). The VLI must indicate the exact length of the Track 2 data actually transmitted.
 It contains the Track 2 image excluding the start sentinel, end sentinel and LRC characters.

Data Field 37**RETRIEVAL REFERENCE NUMBER (RRN)**

Length of Field: 12 bytes, fixed
 Field Type: Alphanumeric (upper case)
 Field Format: Hexadecimal
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
---	M	---	M	M	M	M	M	---	O	---	O	---	O

Description: This data field contains the Retrieval Reference Number (RRN), which is an "ISSwitch Server"-assigned, unique, 12 character reference number used to identify a specific transaction for further processing (e.g., adjustment, void, etc.).
 This entry is shown in hexadecimal format and mapped directly as eight bits per byte, with the value for any binary byte of data varying from hexadecimal "00" to hexadecimal "FF". For example, the 12-byte RRN "1234567890AB" would be transmitted as:

Encoded HEX Bytes												
BYTE	1	2	3	4	5	6	7	8	9	10	11	12
ASCII VALUE	1	2	3	4	5	6	7	8	9	0	A	B
HEX CONTENT	31	32	33	34	35	36	37	38	39	30	41	42

Data Field 38**AUTHORIZATION IDENTIFICATION RESPONSE**

Length of Field: 6 bytes, fixed
Alphanumeric (upper case), left justified, character space filled

Field Type: Hexadecimal

Field Format: Hexadecimal

Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
---	C	---	C	M	---	M	---	---	C	---	---	---	---

Description:

The content of this data field is dependent on the value in Data Field 39, Response Code; and it contains the six-digit approval code for the transaction (On approved transactions).

See the following additional details:

- Sale Completion or Adjustment — It is assumed that this approval code is the same as that for a prior authorization.
- Void of a Transaction — This data field contains the approval code from the original transaction.
- Offline Sale — This data field contains the approval code entered by the Merchant given by a Phone auth system (case of Voice Referral).

This entry is shown in hexadecimal format and mapped directly as eight bits per byte, with the value for any binary byte of data varying from hexadecimal “00” to hexadecimal “FF”. For example, the six-byte Authorization Code “123456” would be transmitted as:

Encoded HEX Bytes						
BYTE	1	2	3	4	5	6
ASCII VALUE	1	2	3	4	5	6
HEX CONTENT	31	32	33	34	35	36

Data Field 39**RESPONSE CODE**

Length of Field: 2 bytes, fixed
Field Type: Alphanumeric (upper case)
Field Format: Hexadecimal
Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
---	M	---	M	---	M	---	M	---	M	---	M	---	M

Description:

This data field contains the Response Code, indicating the American Express disposition for this transaction.

This entry is shown in hexadecimal format and mapped directly as eight bits per byte, with the value for any binary byte of data varying from hexadecimal "00" to hexadecimal "FF". For example, the two-byte Response Code "00" would be transmitted as:

Encoded HEX Bytes		
BYTE	1	2
ASCII VALUE	0	0
HEX CONTENT	30	30

See the Response Code table below.

Data Field 39 ---- Response Code Table

Rsp Code	Action
00	Approval/Accepted
01	Referral / Please Call Issuer
02	Referral / Please Call Issuer
03	Bad SE (Merchant) Number
04	Denied, Pickup Card
05	Do Not Honor / Transaction Declined
10	Partial Approval
12	Invalid transaction
13	Invalid amount
14	Bad Cardholder account information
19	Re-enter Transaction
21	No action Taken
25	Authorization error / Amex:Terminal inactive
30	Format error
31	Error / Bank Not Supported by Switch
41	Lost card
43	Stolen card
45	Product Invalid for Quota Processing
46	Exceed Maximum Quotas
51	Insufficient funds
55	Incorrect PIN
57	Transaction not allowed to Cardholder
58	Transaction not allowed to Terminal
61	Activity/Velocity Amount Exceeded
65	Activity/Velocity Count Exceeded
76	Invalid product
77	Close batch error / Reconciliation Error
78	Conciliation error
80	Conciliation error
82	TRACK2 CVV Error
89	Invalid Merchant/Terminal number
91	Inoperative Switch
94	Duplicate transaction
95	Bad reconciliation - Send detail
96	System Malfunction / Invalid Message
N7	CVV2 error
F1	Terminal Activity Amount Exceeded
F2	Terminal Activity Count Exceeded

F3	Card Activity Amount Exceeded
F4	Card Activity Count Exceeded
F5	Transaction Over Max Amount
F6	Transaction Under Min Amount

Data Field 41**CARD ACCEPTOR TERMINAL IDENTIFICATION**

Length of Field: 8 bytes, fixed
Field Type: Alphanumeric (upper case) & special characters
Field Format: Hexadecimal
Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
M	M	M	M	M	M	M	M	M	M	M	M	M	M

Description:

This data field contains the Card Acceptor Terminal Identification, which is the Acquirer-assigned, eight-character, code that identifies a specific terminal or POS device at a Merchant location. This entry is shown in hexadecimal format and mapped directly as eight bits per byte, with the value for any binary byte of data varying from hexadecimal "00" to hexadecimal "FF". For example, the eight-byte Card Acceptor Terminal ID "12345678" would be transmitted as:

Encoded HEX Bytes								
BYTE	1	2	3	4	5	6	7	8
ASCII VALUE	1	2	3	4	5	6	7	8
HEX CONTENT	31	32	33	34	35	36	37	38

Data Field 42**CARD ACCEPTOR IDENTIFICATION CODE**

Length of Field: 15 bytes, fixed

Field Type: Alphanumeric & special characters, left justified, character space filled

Field Format: Hexadecimal

Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
M	---	M	---	M	---	M	M	M	---	M	---	M	---

Description: This data field contains the Card Acceptor Identification Code, which identifies the Merchant in a POS transaction. Typically, this data field contains the 15 or less digits.

This entry is shown in hexadecimal format and mapped directly as eight bits per byte, with the value for any binary byte of data varying from hexadecimal "00" to hexadecimal "FF". For example, the 10-byte SE Number "1234567890ABCDE" would be transmitted as:

Encoded HEX Bytes															
BYTE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
ASCII VALUE	1	2	3	4	5	6	7	8	9	0	A	B	C	D	E
HEX CONTENT	31	32	33	34	35	36	37	38	39	30	41	42	43	44	45

Data Field 45**TRACK 1 DATA**

Length of Field: 2 bytes minimum, 77 bytes maximum, (LLVAR)
 Variable Len. Indicator: 1 byte BCD, right justified, zero filled
 Field Type: Alphanumeric (upper case) & special characters
 Field Format: Binary Coded Decimal (BCD) and hexadecimal
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	---	C	---	---	---	---	---	---	---	---	---	---	---

Description: This data field contains the information encoded in a valid Track 1 read, preceded by a two-digit, Variable Length Indicator (VLI). The actual Track 1 data is composed of alphanumeric and special characters, and a data field separator value.

In the following example, the two-digit VLI is "65", which is transmitted in binary coded decimal (BCD) format; and the characters that follow are the 65 bytes of ISO 7813 Track 1 data, which are shown in hexadecimal. The caret symbol (^) is used to depict data field separators, and tildes (~) represent character spaces. The total length (VLI plus variable data) is 66 bytes.

B4261449635311004^JOHNSON/JOE~~JR^2701101111610500000000126000000

Encoded HEX Bytes															
BYTE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
ASCII VALUE	`	B	4	2	6	1	4	4	9	6	3	5	3	1	1
HEX CONTENT	60	42	34	32	36	31	34	34	39	36	33	35	33	31	31

BYTE	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
ASCII VALUE	0	0	4	^	J	O	H	N	S	O	N	/	J	O	E
HEX CONTENT	30	30	34	5E	4A	4F	48	4E	53	4F	4E	2F	4A	4F	45

BYTE	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
ASCII VALUE	~	~	J	R	^	2	7	0	1	1	0	1	1	1	1
HEX CONTENT	7E	7E	4A	52	5E	32	37	30	31	31	30	31	31	31	31

BYTE	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
ASCII VALUE	6	1	0	5	0	0	0	0	0	0	0	0	1	2	6
HEX CONTENT	36	31	30	35	30	30	30	30	30	30	30	30	31	32	36

BYTE	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
ASCII VALUE	0	0	0	0	0	0									
HEX CONTENT	30	30	30	30	30	30									

Data Field 52**PIN (Personal Identification Number) Data**

Length of Field: 8 bytes, Fixed
Field Type: 64 N, bit string; 8 bytes
Field Format: Hexadecimal Fixed Length
Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	---	C	---	---	---	---	---	---	---	---	---	---	---

Description: Field 52 contains a PIN, encrypted, and formatted as a block of 16 hexadecimal digits.

A PIN password is a number assigned to a cardholder for unique identification at the point of service or ATM.

This field format must conform to the PIN Block Format Code 0.

A customer PIN must never be logged, even if it is in an encrypted form.

Data Field 54**Additional Amounts**

Length of Field: 4 bytes minimum, 14 bytes maximum, (LLLVAR)

Variable Len. Indicator: 2 byte BCD, right justified, zero filled

Field Type: Numeric, right justified, zero filled

Field Format: Binary coded decimal (BCD) and hexadecimal

Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	---	C	---	C	---	C	---	O	---	---	---	---	---

Description: The Additional Amounts field contains other amounts associated with the transaction. When processing restaurant transactions this field contains the tip. Although this is a variable length data field, the variable data in this entry is always 12 bytes. See the following example

- The three-digit VLI is transmitted in binary coded decimal (BCD) format, and the odd length must be padded. Specifically, "012", must be padded with a leading zero to create the even-length value "00 12", and four digits of data are actually transmitted.
- The 12-digit, fixed length, Additional Amount is right justified and zero filled. The example shown is the entry for a 5.00 tip amount with two decimal places implied.

VLI	AMOUNT DATA	Data to send
012	000000000500	0x0012h'000000000500'

Encoded BCD / HEX Bytes														
BYTE	1	2	3	4	5	6	7	8	9	10	11	12	13	14
ASCII VALUE	VLI		0	0	0	0	0	0	0	0	0	5	0	0
HEX CONTENT	00	12	30	30	30	30	30	30	30	30	30	35	30	30

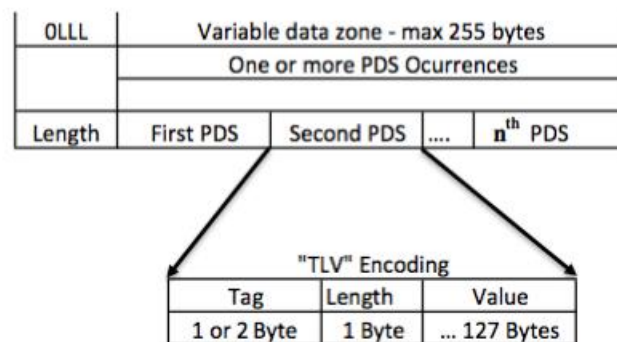
Data Field 55**INTEGRATED CIRCUIT CARD SYSTEM RELATED DATA**

Length of Field: 3 bytes minimum, (LLLVAR) 258 bytes maximum — EMV
 Variable Len. Indicator: 2 byte BCD, right justified, zero filled
 Field Type: Alphanumeric (upper case) & special characters
 Field Format: Binary coded decimal (BCD) and hexadecimal
 Field Requirement:

0100	0110	0200	0210	0220	0230	0320	0330	0400	0410	500	0510	0800	810
C	C	C	C	C	---	C	---	M	---	---	---	---	---

Description: ICC System Related Data (DE 055) is used to transport chip-specific data over a network. It is used for Chip Full Grade transactions whether the ICC data is EMV compliant and in e-commerce transactions where the ICC data may be EMV compliant or issuer specific.

- For issuer specific ICC data, the data zone of DE 055 is considered as a block of binary data with no specific format.
- For EMV compliant ICC data, DE 055 has its own generic structure and may contain one or more Private Data Sub-elements (PDS's), as shown in the figure below.
- PDS's can appear in any order in DE 055



Structure of DE 055, containing PDS's with 'TLV' encoding

Each PDS within DE 055 consists of three sub-fields:

1. Tag: Contains the tag identifying the EMV data object transported in this PDS (e.g. PDS '9F26'). The PDS Tag consists of either one or two bytes, represented as hexadecimal. The second byte is provided only if the continuation indicator in the first byte is set. Refer to the EMV 98 Specifications for the structure and definition of the EMV tags and definition of the continuation indicator.
2. Length: Specifies the length (in bytes) of the PDS data, expressed as a binary number in the range 1 to 127.

3. Value: Contains the actual data from the corresponding EMV data object, as identified by the PDS Tag.

Data Field 55 EMV PDS / Tags

Tag	Req	Rsp	Type	Bytes	Name	Description
0LLL	M	M	BCD	2	DATA LENGTH	BCD length of data to follow, maximum 255 bytes.
5F2A	M	O	BCD	2	Transaction Currency Code	ISO currency code for this transaction. Example: "840" (U.S. Dollars) is entered as "08 40" in 2-byte, BCD format.
71 & 72	---	O	HEX	...128	Issuer Scripts	Issuer Script Template 1 – Tag '71' & '72'- (Response Message) Field 55 may contain zero or more type 71 and type 72 templates, in any order, up to a collective length of 128 bytes.
82	M	O	HEX		Application Interchange Profile (AIP)	Bitmap that indicates ability of the card to support specific functions. The terminal attempts to execute only those functions that the ICC supports.
84	O	O	HEX	...16	Application Identifier (AID) / Dedicated File (DF) Name	Taken from the application (application specific data)
91	---	O	HEX	16	Issuer Authentication Data	Sent by the issuer if on-line issuer authentication is required.
95	M	O	HEX	5	Terminal Verification Result (TVR)	Status of the different functions as seen by the terminal during the processing of a transaction.
9A	M	O	BCD	3	Transaction Date	Terminal-generated Transaction Date, in format "YY MM DD". Example: Jun. 1, 2015 = "150601"
9C	M	O	BCD	16	Transaction Type	Taken from the transaction data. Code indicates type of financial transaction usually given by the first two digits of the ISO 8583 Processing Code.
9F02	M	O	BCD	6	Transaction Amount	Authorization amount of transaction, provided by terminal to the card. Excludes adjustments and could differ from amount received in any alternative Amount data field present in the Authorization Message.
9F03	C	O	BCD	6	Amount Other	A secondary amount associated with the transaction representing a cashback amount. Mandatory for cashback if the amount was entered.

9F09	O	O	HEX	2	Terminal Application Version Number	Taken from the Terminal application (application specific data)
------	---	---	-----	---	-------------------------------------	---

Data Field 55 EMV PDS / Tags (CONTINUED)

Tag	Req	Rsp	Type	Bytes	Name	Description
9F1A	M	O	BCD	2	Terminal Country Code	ISO country code for terminal location. Example: "840" (Mexico) is entered as "08 40" in 2-byte, BCD format.
9F1E	O	O	HEX		Interface Device (IFD) Serial Number	Unique and permanent serial number assigned to the Interface Device by the manufacturer.
9F26	M	O	HEX	8	Application Cryptogram (AC)	The Application Cryptogram generated by the chip card in response to GENERATE AC Command. In an online authorization message, this will be the Authorization Request Cryptogram (ARQC).
9F27	M	O	HEX	16	Cryptogram Information Data (CID)	Indicates type of cryptogram (TC, ARQC or AAC) returned by the card, and actions to be performed by terminal.
9F33	O	O	HEX	3	Terminal Capabilities	Specifies the capabilities of the terminal
9F34	O	O	HEX	3	CVM Results (CVMR)	Result of the last cardholder verification method
9F35	O	O	BCD	1	Terminal Type	Specifies the type of terminal
9F36	M	O	HEX	2	(ATC) Application Transaction Counter	Counter maintained by application on the card. Chip Card increments this value for each transaction. Because counter includes failed transactions, this value cannot be used alone to track last transaction.
9F37	M	O	HEX	4	Unpredictable Number	A terminal-generated Unpredictable Number, which is a randomly generated value that adds variability and uniqueness to the creation of the application cryptogram value in the preceding APPLICATION CRYPTOGRAM subfield.
9F41	O	O	BCD	4	Transaction Sequence Counter	Counter maintained by the terminal that is incremented by one for each transaction
9F53	O	O	HEX	1	Transaction Category Code	Usually provided by the acquirer

5F34	M	O	BCD	1	PAN sequence number	Identifies and differentiates card applications with same Primary Account Number (PAN). Both PAN & PAN Sequence Number are required to validate Application Cryptogram.
------	---	---	-----	---	---------------------	---

Data Field 60

PRIVATE USE DATA

Length of Field: 3 bytes minimum, (LLLVAR)
Variable Len. Indicator: 2 bytes, BCD
Field Type: Alphanumeric (upper case)
Field Format: Binary coded decimal (BCD) and hexadecimal
Field Requirement:

MTI	
0100	---
0110	---
0200	---
0210	---
0220	O
0230	---
0320	M
0330	O
0400	---
0410	---
0500	M
0510	O
0800	O
0810	---

Description:

The content of this data field varies:

- Message Type Identifier (MTI) and Systems Trace Audit Number (STAN) — This data field contains the four-digit MTI that indicates the type of data contained in this message, and the STAN, which is a system-generated sequential number that uniquely identifies a transaction.
- Batch Number — This data field contains the Batch Identification Number assigned by the terminal system when a batch is closed. This value is used when referencing a specific group of financial transactions.
- Original Message Data — This data field contains the Message Type and Systems Trace Audit Number (STAN) from the original transaction.
- Software ID — This data field contains the current software version in use by the terminal.

Data Field 60**PRIVATE USE DATA (Continued)**

MTI	Usage
0220	Original Amount
0320	Original Message Data
0500	Batch Number
0800	Logon Request Data
0800	Software ID

MTI - 0500**BATCH NUMBER**

Field	Length	Field Type	Value
LLL VLI Variable Length Indicator	2 bytes	BCD	0006 BCD length of data to follow
Batch Number	3 bytes	BCD	

MTI - 0320**ORIGINAL MESSAGE DATA**

Field	Length	Field Type	Value
LLL VLI Variable Length Indicator	2 bytes	BCD	0010 BCD length of data to follow
Original Message Type (MTI)	4 bytes	AN	e.g. '0200' ASCII
Original Trace Audit Number	6 bytes	AN	e.g. '000001' ASCII

MTI 0220/0420**ORIGINAL MESSAGE DATA**

Field	Length	Field Type	Value
--------------	---------------	-------------------	--------------

LLL VLI Variable Length Indicator	2 bytes	BCD	0012 BCD length of data to follow
Original Amount	12 bytes	AN	e.g. '000000010100' ASCII meaning 101.00 amount

Data Field 61

PRIVATE USE DATA

Length of Field: 3 bytes minimum, (LLLVAR)
Variable Len. Indicator: 2 bytes, BCD
Field Type: Alphanumeric (upper case)
Field Format: Binary coded decimal (BCD) and hexadecimal
Field Requirement:

MTI	
0100	O
0110	---
0200	O
0210	---
0220	C
0230	---
0320	C
0330	---
0400	---
0410	---
0500	---
0510	---
0800	---
0810	---

Description: PRODUCT CODES
The product codes field is used to carry the descriptive information of the items purchased.
The terminal may deliver a maximum of four product codes per

transaction.

The product codes sent by the terminal varies, depending on the point of sale functionality.

For American Express transactions American Express assigns, two-digit, Product/Item Descriptor Codes, which correspond to the merchandise and/or services sold using this POS terminal or device.

These codes are provided to the merchant during set-up.

**PRIVATE USE DF 61 --
PRODUCT CODES**

Field	Length	Field Type	Value
LLL VLI Variable Length Indicator	2 bytes	BCD	0008 BCD length of data to follow
Product Code 1	2 bytes	ANS	
Product Code 2	2 bytes	ANS	
Product Code 3	2 bytes	ANS	
Product Code 4	2 bytes	ANS	

Length of Field: 3 bytes minimum, (LLLVAR)
Variable Len. Indicator: 2 bytes, BCD
Field Type: Alphanumeric (upper case)
Field Format: Binary coded decimal (BCD) and hexadecimal
Field Requirement:

MTI	
0100	M
0110	---
0200	M
0210	---
0220	C
0230	---
0320	C
0330	---
0400	C
0410	---
0500	---
0510	---
0800	---
0810	---

Description: Invoice Number — A.k.a., Record of Charge or ROC Number (American Express) assigned to this transaction. This value is usually generated by the POS terminal or device for each financial transaction. The same Invoice Number must be used on a Reversal or Void as the Original Transaction.
This entry cannot be “000000” (six zeros), and the value must be unique within a batch.
Although this is a variable length data field, the variable data in this entry is always six bytes. See the following examples:
The three-digit VLI is transmitted in BCD format, and the odd length must be padded. Specifically, “006”, must be padded with a leading zero to create the even-length value “00 06”, and four digits of data for the VLI are actually transmitted.
The Invoice Number is transmitted in hexadecimal format, right justified and zero filled to six bytes.

PRIVATE USE DF 62
(Continued)

Field	Length	Field Type	Value
LLL VLI Variable Length Indicator	2 bytes	BCD	0008 BCD length of data to follow
Invoice/ECR reference number	6 bytes	ANS	Invoice number as entered by the operator or generated automatically by the terminal.

Length of Field: 3 bytes minimum, (LLLVAR)
Variable Len. Indicator: 2 bytes, BCD
Field Type: Alphanumeric (upper case)
Field Format: Binary coded decimal (BCD) and hexadecimal
Field Requirement:

MTI	
0100	M
0110	---
0200	M
0210	---
0220	C
0230	---
0320	C
0330	---
0400	C
0410	---
0500	---
0510	---
0800	---
0810	---

Description: The content of this data field varies:
This data field is used in several messages to transport different information, and its contents are limited by the Message Type Identifier (MTI) entry for this transaction.

RECONCILIATION REQUEST TOTALS

The reconciliation totals are sent to the host in the settlement message (message type 0500). The totals are the capture, debit and authorized totals of the current open batch. In the case of multi-host applications, the totals are only those totals associated with each acquirer.

DF 63

RECONCILIATION REQUEST TOTALS

Field	Position Exc LLL	Length	Field Type	Value	
LLL VLI Variable Length Indicator		2 bytes	BCD	0008 BCD length of data to follow	
Credit Sales Count	1	3 Bytes	AN	000-999	GROUP 1
Credit Sales Amount	4	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Refund Count	16	3 Bytes	AN	000-999	
Refund Amount	19	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Credit Cash Count	31	3 Bytes	AN	000-999	
Credit Cash Amount	34	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Debit Sales Count	46	3 Bytes	AN	000-999	
Debit Sales Amount	49	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Debit Refund Count	61	3 Bytes	AN	000-999	GROUP 2
Debit Refund Amount	64	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Authorization Sales Count	76	3 Bytes	AN	000-999	
Authorization Sales Amount	79	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Authorization Refund Count	91	3 Bytes	AN	000-999	
Authorization Refund Amount	94	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
HYPERCOM RESERVED	106	3 Bytes	AN	000	RESV
HYPERCOM RESERVED	109	12 bytes	AN	000000000000	
Void Sales Count	121	3 Bytes	AN	000-999	GROUP 3
Void Sales Amount	124	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Void Refund Count	136	3 Bytes	AN	000-999	
Void Refund Amount	139	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Payment Count	151	3 Bytes	AN	000-999	GROUP 4
Payment Amount	154	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Payment Count	166	3 Bytes	AN	000-999	
Payment Amount	169	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
Loyalty Count	181	3 Bytes	AN	000-999	GROUP 5
Loyalty Amount	184	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	
CashBack Count	196	3 Bytes	AN	000-999	
CashBack Amount	199	12 bytes	AN	\$\$\$\$\$\$\$\$\$çç	

* Each set of Count/Amount is indicated in distinct colors.

* GROUP 1 is mandatory. If a value of GROUP 2 to GROUP 4 have to be used, all the values of the Group and all the presceding groups must be filled with Zeroes

The Reconciliation Totals are calculated using the following rules:
Only non-Reversed or Voided transactions are summarized. Except when using the Voided Transactions Group 3, where voided transactions are summarized.

Credit Cards are those cards processed using normal processing method with no PIN Entry or Account Selection.

Debit Cards are those cards that are processed using PIN Entry and Account Selection Authorize Cards fields must not be used; they are present for compatibility only.

The Sale Count is a count of all non-Voided transactions that would cause funds to transfer from the cardholder to the merchant.

The Sale Amount is the total amount of all non-Voided transactions that would cause funds to transfer from the cardholder to the merchant

The Refund Count is a count of all non-Voided transactions that would cause funds to transfer from the merchant to the cardholder.

The Refund Amount is the total amount of all non-Voided transactions that would cause funds to transfer from the merchant to the cardholder.

Any transaction that the host receives that has a value of \$0.00 is a Voided transaction, and should not be included in the count.

The terminals will never allow an ADJUST to \$0.00, so there can be no confusion with a Voided transaction.

The authorization totals fields are compatibility fields.

The cash advance totals fields are application dependent. If an application requires these fields, the authorization fields are required, but may be 0 filled.

The loyalty totals fields are application dependent. If an application requires these fields, the authorization fields are required, but may be 0 filled.

The payment totals fields are application dependent. If an application requires these fields, the authorization fields are required, but may be 0 filled.

The Payment Count is a count of all non-Voided card payment transactions that would cause funds to transfer from the merchant to the cardholder.

The Payment Amount is the total amount of all non-Voided card payment transactions that would cause funds to transfer from the merchant to the cardholder.

ADDITIONAL DATA ELEMENTS - TAGS -

The additional data can consist of one or more of the following fields immediately following each other. Each of the fields contains its own length indicator to allow any fields that are not recognized by the host to be stepped over and ignored, with any following fields still being processed successfully.

DF 63		ADDITIONAL DATA ELEMENTS		
Field	Length	Field Type	Value	
LLL VLI Variable				
Length Indicator	2 bytes	BCD	0LLL BCD length of data to follow	
Additional Data (1)	2 bytes	ANS	First additional data element	
Additional Data (2)	2 bytes	ANS	Second additional data element	
:	:	:	The additional data can be any number of data elements in any order. Each data element contains its own identifier and length. Unknown elements may be skipped over.	
:	:	:		
:	:	:		
Additional Data (x)	2 bytes	ANS	Last additional data element	

CVV2 DATA (16)

Field	Length	Field Type	Value
LLL VLI Additional Data			
Length	2 bytes	BCD	0LLL BCD length of data to follow
TAG Id	2 bytes	ANS	'16' CVV2 Data
Request Data	6 bytes	AN	The 6 character CVV2 Request Value sent to the Host Position 1 0 - CVV2 Value is intentionally not provided 1 - CVV2 Value is Present 2 - CVV2 Value is Present but illegible 9 - Cardholder states that no CVV2 value is imprinted Position 2 0 - Only the normal Response Code should be returned 1 - Both the Response Code and the CVV2 Result Code should be returned Positions 3-6 The CVV2 Value as imprinted on the card (right-justify and space fill the entry) If position 1 = 0, 2, or 9, positions 3-6 should be space filled
Result Code (Response Only)	1 bytes	AN	The 1 character CVV2 Result Code returned from the Host M - CVV2 Match N - CVV2 No Match P - Not Processed S - Merchant has indicated that CVV2 is not present on card U - Issuer is not certified

ALTERNATE HOST RESPONSE (22)

Field	Length	Field Type	Value
LLL VLI Additional Data Length	2 bytes	BCD	0042 BCD length of data to follow
TAG Id	2 bytes	ANS	'22' Alternate Host Response indicator
Alternate Host Response	... 25 bytes	ANS	Alternate text to display as the host response

ADDITIONAL HOST PRINT DATA (29)

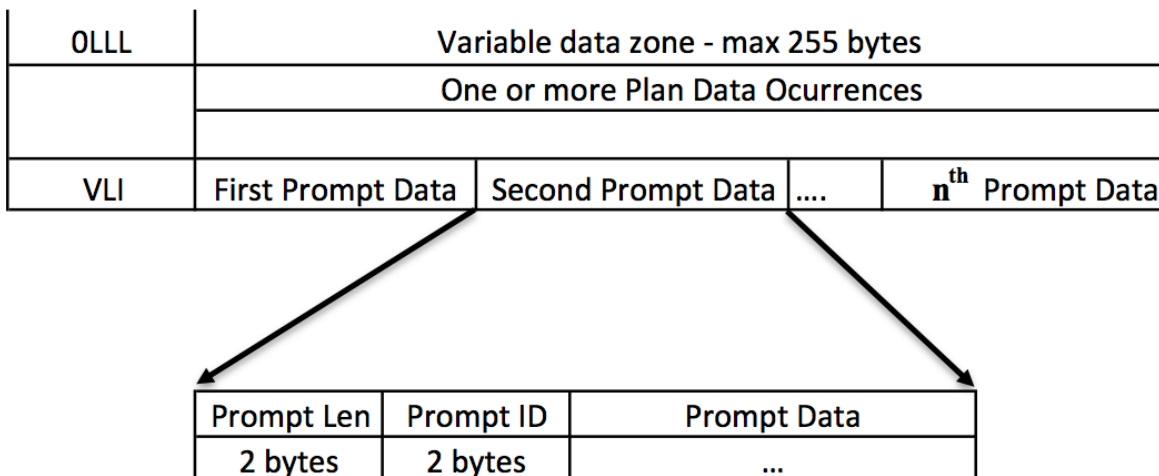
Field	Length	Field Type	Value
LLL VLI Additional Data Length	2 bytes	BCD	0LLL BCD length of data to follow
TAG Id	2 bytes	ANS	'29' Alternate Host Response indicator
Alternate Host Response	... 255 bytes	ANS	Variable length print data. Data may include carriage return (CR), linefeed (LF), (09h) tab and ASCII characters between 20h and 7Ah. LF and CR behave in the same fashion.

ADDITIONAL PROMPT DATA (35)

Field	Length	Field Type	Value
LLL VLI Additional Data Length	2 bytes	BCD	0LLL BCD length of data to follow
TAG Id	2 bytes	ANS	'35' Additional Prompt Data
Additional Prompts Data	... 255	ANS	Additional Prompts Data consists of a series of 1 or more instances of the following Record with three subfields:

ADDITIONAL PROMPT DATA

SubField	Len	Value
Prompt Length	2	2 digits decimal
Prompt ID	2	2 digits alpha
Prompt Data	...	Prompt Data Length is (Prompt Length -2). The Prompt ID is not counted when calculating the Prompt Length



TAX AMOUNT #1 (39)

Field	Length	Field Type	Value
LLL VLI Additional Data			
Length	2 bytes	BCD	0014 BCD length of data to follow
TAG Id	2 bytes	ANS	'39' Tax Amount #1
Tax Amount	12 bytes	ANS	0000000000v00 - 9999999999v99 Where the last 2 digits are the decimals (i.e., 10 digits, plus 2 decimals). ("v" indicates implied decimal place. Do not include 'v' or "." in actual entry).

TAX AMOUNT #2 (40)

Field	Length	Field Type	Value
LLL VLI Additional Data			
Length	2 bytes	BCD	0014 BCD length of data to follow
TAG Id	2 bytes	ANS	'40' Tax Amount #2
Tax Amount	12 bytes	ANS	0000000000v00 - 9999999999v99 Where the last 2 digits are the decimals (i.e., 10 digits, plus 2 decimals). ("v" indicates implied decimal place. Do not include 'v' or "." in actual entry).

CASHBACK AMOUNT (41)

Field	Length	Field Type	Value
LLL VLI Additional Data			
Length	2 bytes	BCD	0014 BCD length of data to follow
TAG Id	2 bytes	ANS	'41' Cash Back Amount

Tax Amount

12
bytes

ANS

0000000000v00 - 9999999999v99
Where the last 2 digits are the
decimals (i.e., 10 digits, plus 2
decimals).
("v" indicates implied decimal
place.
Do not include 'v' or "." in actual
entry).

PAYMENT PLAN (45)

Field	Length	Field Type	Value
LLL VLI Additional Data	2		0008 BCD length of data to
Length	bytes	BCD	follow
TAG Id	2	ANS	'45' Payment Plan
	bytes		
Number of Payments	2	AN	Number of payments selected at
	bytes		the terminal
Payment Plan	4	ANS	The payment plan selected at the
	bytes		terminal
Transaction Type	1 byte	ANS	Used in Payment Plan Query in
			the Balance Transaction
			'S' Querying available data for a
			Sale Trans
			'C' Querying available data for a
			Cash Trans
			Blank for Sale and Cash
			Transactions

PAYMENT PLAN (45) – Continued – Sample Values

REQUEST VALUES

Payment Plan Query Request

```
00000000000000000000000000000000S
**++++-----*
|  |  |  |
|  |  |  | +> Transaction Type ( C = Extra financing , S = Quota Sale )
|  |  |  |
|  |  |  | +> Fill
|  |  |  |
|  |  |  |
|  |  |  |
|  |  |  | +> Plan ID ( 0000 for Plan Query )
|  |  |  |
|  |  |  | +> Number of Quotas ( 00 for Plan Query )
```

Payplan Sale/Cash Request

```
03EXT10000000000000000000000000C
**++++-----*
|  |  |  |
|  |  |  | +> Transaction Type ( C = Extra financing , S = Quota Sale )
|  |  |  |
|  |  |  |
|  |  |  | +>Interest Rate (2 integers, 2 decimals )
|  |  |  |
|  |  |  | +> Plan ID
|  |  |  |
|  |  |  | +> Number of Quotas
```

PAYMENT PLAN (45) – Continued – Sample Values

RESPONSE VALUES

Payment Plan Response for Extra financing

```
03EXT100150000000001005000000100101
*****
|   |   |   |   |
|   |   |   |   | +> Extra financing Final Value
|   |   |   |   |
|   |   |   |   | +> Quota Amount
|   |   |   |   |
|   |   |   |   | +>Interest Rate (2 integers, 2 decimals )
|   |   |   |   |
|   |   |   |   | +> Plan ID
|   |   |   |   |
+> Number of Quotas
```

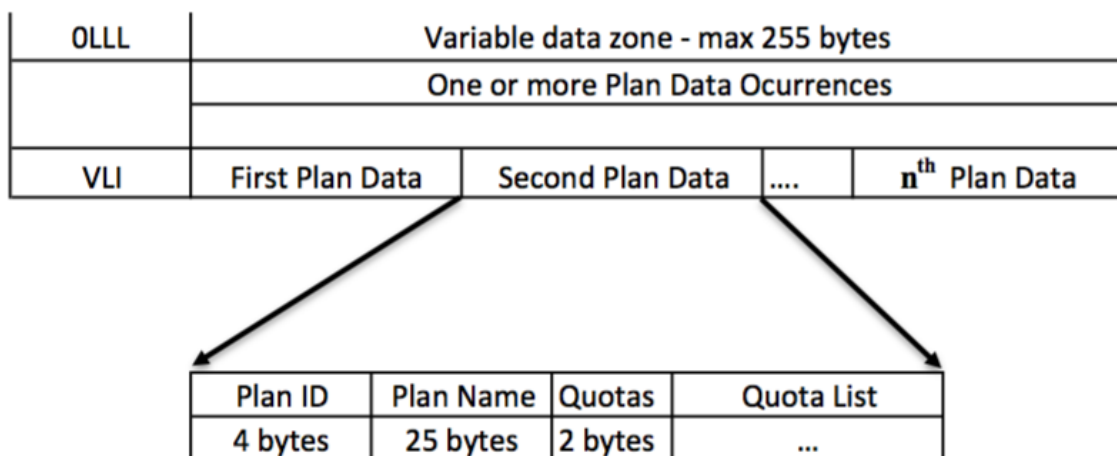
PAYMENT PLAN QUERY RESPONSE (46)

Field	Length	Field Type	Value
LLL VLI Additional Data			
Length	2 bytes	BCD	0LLL BCD length of data to follow
TAG Id	2 bytes	ANS	'46' Payment Plan Query Response
Payment Plan DATA	... 255	ANS	Payment Plan Data consists of a series of 1 or more instances of the following Record with three subfields:

PAYMENT PLAN DATA

SubField	Len	Value
Plan ID	4	4 chars alphanumeric
Plan Name	25	25 chars left justified space filled
Nbr of Quotas	2	Number of two digit quotas to follow
Quota List	...	

PAYMENT PLAN QUERY RESPONSE (46) - CONTINUED



The following example an example of how to parse this data.
Given the data:

"INT1INTRA FINANCING 020609EXT1EXTRA FINANCING 03061224"

[illegible]

ORIGINAL TAX AMOUNT #1 (82)

Field	Length	Field Type	Value
LLL VLI Additional Data			
Length	2 bytes	BCD	0014 BCD length of data to follow
TAG Id	2 bytes	ANS	'82' Tax Amount #1

Tax Amount	12 bytes	ANS	0000000000v00 - 9999999999v99 Where the last 2 digits are the decimals (i.e., 10 digits, plus 2 decimals). ("v" indicates implied decimal place. Do not include 'v' or "." in actual entry). When processing in Honduras, by local laws, an 8% discount must be applied on the Tax Amount given by the Merchant. Original Tax - 8% must be sent on Tag 39. Original Tax amount has to be sent on this tag 82
------------	-------------	-----	---

ACQUIRER ENTRY CAPABILITIES (1C)

Field	Length	Field Type	Value
LLL VLI Additional Data	2		
Length	bytes	BCD	0001 BCD length of data to follow
TAG Id	2	ANS	'1C' ACQUIRER ENTRY CAPABILITIES
ENTRY CAPABILITIES BITMAP	bytes	ANS	Entry Capabilities are represented in a Bitmap of One Byte. This Byte should be sent as an Hexadecimal representation of the value.
	2		Bits. 12345678 + RFU + RFU + RFU + RFU + RFU + EMV Entry Capability + Magstripe Swiped + Manual Keyed Entry
If the Acquirer processing in the Terminal only has Magnetic Stripe and Keyed Entry capabilities 11000000 -> represented in Hexadecimal C0			If the Acquirer processing in the Terminal has Chip Card ,Magnetic Stripe and Keyed Entry capabilities 11100000 -> represented in Hexadecimal E0
006000431434330 +++++			006000431434530 +++++
+ TAG VALUE C0			+ TAG VALUE E0
+ TAG 1C			+ TAG 1C
+ TAG LLVAR			+ TAG LLVAR
+ DE 63 LLLVAR			+ DE 63 LLLVAR

*** A merchant terminal could have one or more different Acquirers (Brands) and the Terminal must be prepared to send distinct values in this TAG. E.g. having VISA EMV certified (value E0) and having MasterCard with only Magstripe (value C0).

6.0 Transaction Processing Considerations

6.1 Payment Plan Transactions: Extra financing / Quota Sale

The Extra financing or Quota Transactions are handled as a normal transaction transaction adding specific information in Data Element 63, using private Data Element Tag 45 the information handled this field indicates the selected quotas/months and type of plan used. The plan ID is an alphanumeric four-digit value; the quota is given in a two-digit value 00-99.

The available Plans and the quotas for each of them is dynamically loaded before the Sale/Cash transaction happens. The terminal has to query “ISSwitch Server” the available data using a Balance (0100/31xxxx) transaction, adding the private data tag 45 of the data element 63 to identify the Balance as a “Plans Query” transaction. The response message (0110) to the balance transaction if approved (DE 39 = 00), will contain the private data tag 46, containing the detail of the available Plans and Quotas.

When processing with Visa Argentina as endpoint, in general, we have two types of Pay Plan Transactions, Intra-financing or Quota Sale and Extra-Financing.

Quota Sale transactions are represented as Sale transactions, this transaction need some special data to be added to be identified as an Intra-financing or quota sale transaction: a plan ID and the number of payments or number of quotas.

The Extra-financing transactions are handled as Cash Advance transactions, as the quota sale, the Plan ID and the number of payments are needed.

The Plan ID and Number of payments available are queried to the server to be displayed and selected on the merchant terminal.

When querying for Quota Sale financing available plans, the value “S” must be used on the subfield “Transaction Type” of private data tag 45 on the request message.

When querying for Extra-financing available plans, the value “C” must be used on the subfield “Transaction Type” of private data tag 45 on the request message.

6.1.1 Payplan Query Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	01	00	20	20	05	80	20	C0	00	02	31
17	00	00	00	00	40	00	22	00	03	00	33	54	13	33	00	89
33	02	01	02	D2	51	22	01	05	05	90	10	3F	30	30	30	30
49	30	30	30	31	30	30	30	30	30	30	30	30	30	30	30	30
65	30	30	31	00	49	00	25	34	35	30	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	53
97	00	04	31	43	43	30										
113																

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	01H	00H	20H	20H	05H	80	20H	C0	00H	02H	1
17	00H	00H	00H	00H	@	00H	"	00H	03H	00H	3	T	13H	3	00H	89
33	02H	01H	02H	D2	Q	"	01H	05H	05H	90	10H	?	0	0	0	0
49	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
65	0	0	1	00H	I	00H	%	4	5	0	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	S
97	00H	04H	1	C	C	0										
113																

Parsed Message

```

TPDU: 6000030000
MsgType:0100
BitMap:0010000000010000000000010110000000000100000011000000000000000000000000010
003=310000
011=000040
022=022
024=0003
025=00
035=5413330089020102D2512201050590103
041=00000001
042=000000000000000001
063= TAGS
      45=00000000000000000000000000000000S
      1C=C0

```

6.1.2 Payplan Query Response Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	03	01	10	70	3C	01	80	0E	80	00	02	16
17	54	13	33	00	89	02	01	02	31	00	00	00	00	00	00	00
33	00	00	00	40	10	02	14	05	13	25	12	00	03	00	36	31
49	33	34	31	30	30	30	30	30	34	30	51	55	4F	54	30	30
65	30	30	30	30	30	30	30	30	30	31	01	11	00	25	34	35
81	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
97	30	30	30	30	30	30	53	00	04	31	43	43	30	00	76	34
113	36	45	58	54	31	45	58	54	52	41	20	46	49	4E	41	4E
129	43	49	4E	47	20	20	20	20	20	20	20	20	20	20	30	33
145	30	33	30	36	31	32	49	4E	54	31	49	4E	54	52	41	20
161	46	49	4E	41	4E	43	49	4E	47	20	20	20	20	20	20	20
177	20	20	20	30	33	30	33	30	36	30	39					

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	03H	01H	10H	p	<	01H	80	0EH	80	00H	02H	16H
17	T	13H	3	00H	89	02H	01H	02H	1	00H	00H	00H	00H	00H	00H	00H
33	00H	00H	00H	@	10H	02H	14H	05H	13H	%	12H	00H	03H	00H	6	1
49	3	4	1	0	0	0	0	0	4	0	Q	U	O	T	0	0
65	0	0	0	0	0	0	0	0	0	1	01H	11H	00H	%	4	5
81	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
97	0	0	0	0	0	0	S	00H	04H	1	C	C	0	00H	v	4
113	6	E	X	T	1	E	X	T	R	A	20H	F	I	N	A	N
129	C	I	N	G	20H	20H	20H	20H	20H	20H	20H	20H	20H	20H	0	3
145	0	3	0	6	1	2	I	N	T	1	I	N	T	R	A	20H
161	F	I	N	A	N	C	I	N	G	20H	20H	20H	20H	20H	20H	20H
177	20H	20H	20H	0	3	0	3	0	6	0	9					

Parsed Message

TPDU: 6000000003
MsgType:0110
BitMap:011100000011110000000000110000000000011101000000000000000000000010
002=5413330089020102
003=310000
004=000000000000
011=000040
012=100214
013=0513
014=2512
024=0003
025=00
037=613410000040
038=QUOT00
039=00
041=00000001
063=TAGS
45=00000000000000000000000000000000S
1C=C0
46=EXT1EXTRA FINANCING 03030612INT1INTRA FINANCING 03030609

6.1.3 Payplan Sale Request Sample Transaction

HEXADECIMAL REPRESENTATION

[illegible]

ASCII REPRESENTATION

[illegible]

Parsed Message

```
TPDU: 6000030000  
MsgType:0200  
BitMap:00110000000111000000000101100000000010100011000000000000000000000110  
003=000000  
004=0000000010120  
011=600025  
012=114503  
013=0530  
022=022  
024=0003  
025=00  
035=5413330089020102D2512201050590103  
037=615111600024  
041=00000001  
042=0000000000000001  
062=000022  
063=TAGS  
    40=0000000000120  
    45=06INT10000000000000000S
```

6.1.4 Payplan Sale Response Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	03	02	10	70	3C	01	80	0E	80	00	00	16
17	54	13	00	00	00	00	01	02	00	00	00	00	00	00	01	01
33	20	60	00	25	11	45	03	05	30	25	12	00	03	00	36	31
49	35	31	31	31	36	30	30	30	32	34	34	35	32	33	34	32
65	30	30	30	30	30	30	30	30	30	31						

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	03H	02H	10H	p	<	01H	80	0EH	80	00H	00H	16H
17	T	13H	00H	00H	00H	00H	01H	02H	00H	00H	00H	00H	00H	00H	01H	01H
33	20H	`	00H	%	11H	E	03H	05H	0	%	12H	00H	03H	00H	6	1
49	5	1	1	1	6	0	0	0	2	4	4	5	2	3	4	2
65	0	0	0	0	0	0	0	0	0	1						

Parsed Message

TPDU: 6000000003
 MsgType:0210
 BitMap:011100000011110000000000110000000000011101000000000000000000000000
 002=5413000000000102
 003=000000
 004=000000010120
 011=600025
 012=114503
 013=0530
 014=2512
 024=0003
 025=00
 037=615111600024
 038=452342
 039=00
 041=00000001

6.1.5 Payplan Cash/Extra-financing Request Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	30	38	05	80	28	C0	00	06	01
17	00	00	00	00	00	00	20	00	60	00	21	13	45	23	05	27
33	00	22	00	03	00	33	54	13	33	00	89	02	01	02	D2	51
49	22	01	05	05	90	10	3F	36	31	34	38	31	33	36	30	30
65	30	32	30	30	30	30	30	30	30	30	31	30	30	30	30	30
81	30	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30
97	32	30	00	43	00	25	34	35	30	33	45	58	54	31	30	30
113	30	30	30	30	30	30	30	30	30	30	30	30	30	30	43	

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	0	8	05H	80	(	C0	00H	06H	01H
17	00H	00H	00H	00H	00H	00H	20H	00H	`	00H	!	13H	E	#	05H	'
33	00H	"	00H	03H	00H	3	T	13H	3	00H	89	02H	01H	02H	D2	Q
49	"	01H	05H	05H	90	10H	?	6	1	4	8	1	3	6	0	0
65	0	2	0	0	0	0	0	0	0	0	1	0	0	0	0	0
81	0	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0
97	2	0	00H	C	00H	%	4	5	0	3	E	X	T	1	0	0
113	0	0	0	0	0	0	0	0	0	0	0	0	0	0	C	

Parsed Message

TPDU: 6000030000
 MsgType: 0200
 BitMap: 0011000000111000000001011000000000101000110000000000000000000000110
 003=010000
 004=000000002000
 011=600021
 012=134523
 013=0527
 022=022
 024=0003
 025=00
 035=5413330089020102D2512201050590103
 037=614813600020
 041=00000001
 042=0000000000000001
 062=000020
 063=TAGS
 45=03EXT100000000000000000C

6.1.6 Payplan Cash/Extra-financing Response Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	03	02	10	70	3C	01	80	0E	80	00	02	16
17	54	13	33	00	89	02	01	02	01	00	00	00	00	00	00	20
33	00	60	00	21	13	45	23	05	27	25	12	00	03	00	36	31
49	34	38	31	33	36	30	30	30	32	30	34	35	33	34	32	33
65	30	30	30	30	30	30	30	30	30	31	00	38	00	36	34	35
81	30	33	45	58	54	31	30	30	31	35	30	30	30	30	30	30
97	30	30	31	30	30	35	30	30	30	30	30	30	31	30	30	31
113	30	31														

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	03H	02H	10H	p	<	01H	80	0EH	80	00H	02H	16H
17	T	13H	3	00H	89	02H	01H	02H	01H	00H	00H	00H	00H	00H	00H	20H
33	00H	`	00H	!	13H	E	#	05H	'	%	12H	00H	03H	00H	6	1
49	4	8	1	3	6	0	0	0	2	0	4	5	3	4	2	3
65	0	0	0	0	0	0	0	0	0	1	00H	8	00H	6	4	5
81	0	3	E	X	T	1	0	0	1	5	0	0	0	0	0	0
97	0	0	1	0	0	5	0	0	0	0	0	0	1	0	0	1
113	0	1														

Parsed Message

```

TPDU: 6000000003
MsgType:0210
BitMap:01110000001111000000000011000000000000111010000000000000000000000010
002=5413330089020102
003=010000
004=000000002000
011=600021
012=134523
013=0527
014=2512
024=0003
025=00
037=614813600020
038=453423
039=00
041=00000001
063=TAGS
45=03EXT100150000000001005000000100101

```


6.2 Loyalty (Miles / Points) Transactions

6.2.1 Definition 1 Miles / Points Redemption

~~The Loyalty transactions must be summarized at Group 5 of the Settlement Totals~~

~~ATTN: WORK IN PROGRESS. TO BE COMPLETED~~

6.2.2 Definition 2 Financial Transactions applied to Points/Miles

The Points/Miles transactions when handled as financial transactions, are transported as regular Sale/Cash Transactions, but adding some specific values to DE 3 to identify this transaction as a Loyalty transaction and the funds has to be taken from the Loyalty balance.

Data Element 3, positions 2-3 must be set to value “95”, to identify the source of funds as Loyalty.

Normal tax and tip processing could be applied to these transactions.

If the transaction is approved (DE 39 = 00), the merchant terminal could receive in response message 0210 the private tag 53 Misc., Amount on DE 63, indicating the Points/Miles applied to process the transaction.

The Financial transactions applied to the Loyalty Balance, must be summarized as normal transactions Sales / Cash in Settlement Totals.

6.2.2.1 Loyalty Sale Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	05	00	00	02	00	30	20	05	80	20	C0	00	06	00
17	95	00	00	00	00	00	10	12	60	00	01	00	22	00	05	00
33	37	44	44	44	44	44	44	44	44	D1	60	81	01	25	99	36
49	84	00	00	0F	37	30	31	30	30	30	39	34	37	30	31	30
65	30	30	30	30	30	30	30	30	30	39	34	00	06	30	30	30
81	30	30	36	00	38	00	14	33	39	30	30	30	30	30	30	30
97	30	30	30	31	32	00	14	38	32	30	30	30	30	30	30	30
113	30	30	30	31	33	00	04	31	43	45	30					

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	05H	00H	00H	02H	00H	0	20H	05H	80	20H	C0	00H	06H	00H
17	95	00H	00H	00H	00H	00H	10H	12H	`	00H	01H	00H	"	00H	05H	00H
33	7	D	D	D	D	D	D	D	D	D1	`	81	01H	%	99	6
49	84	00H	00H	0FH	7	0	1	0	0	0	9	4	7	0	1	0
65	0	0	0	0	0	0	0	0	0	9	4	00H	06H	0	0	0
81	0	0	6	00H	8	00H	14H	3	9	0	0	0	0	0	0	0
97	0	0	0	1	2	00H	14H	8	2	0	0	0	0	0	0	0
113	0	0	0	1	3	00H	04H	1	C	E	0					

Parsed Message

TPDU: 6000050000

MsgType:0200

BitMap:0011000000100000000001011000000000100000110000000000000000000000110

003=009500

004=000000001012

011=600001

022=022

024=0005

025=00

035=44444444444444444444D16081012599368400000

041=70100094

042=7010000000000094

062=000006

063=TAG

39=000000000012

82=000000000013

1C=E0

6.2.2.2 Loyalty Sale Response Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	05	02	10	70	3C	01	80	0E	80	00	02	16
17	44	44	44	44	44	44	44	44	00	95	00	00	00	00	00	10
33	12	60	00	01	09	45	04	05	26	16	08	00	05	00	30	30
49	30	30	37	30	30	39	34	35	30	34	30	34	30	38	34	34
65	30	30	37	30	31	30	30	30	39	34	00	16	00	14	35	33
81	30	30	30	30	30	30	30	30	30	30	33	35				

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	05H	02H	10H	p	<	01H	80	0EH	80	00H	02H	16H
17	D	D	D	D	D	D	D	D	00H	95	00H	00H	00H	00H	00H	10H
33	12H	`	00H	01H	09H	E	04H	05H	&	16H	08H	00H	05H	00H	0	0
49	0	0	7	0	0	9	4	5	0	4	0	4	0	8	4	4
65	0	0	7	0	1	0	0	0	9	4	00H	16H	00H	14H	5	3
81	0	0	0	0	0	0	0	0	0	0	3	5				

Parsed Message

TPDU: 6000000005
 MsgType:0210
 BitMap:0111000000111100000000001100000000000111010000000000000000000000010
 002=4444444444444444
 003=009500
 004=000000001012
 011=600001
 012=094504
 013=0526
 014=1608
 024=0005
 025=00
 037=000070094504
 038=040844
 039=00
 041=70100094
 063=TAGS
 53=000000000035

6.2.2.3 Loyalty Balance Request Sample Transaction

HEXADECIMAL REPRESENTATION

[illegible]

ASCII REPRESENTATION

[illegible]

Parsed Message

```
TPDU: 6000050000
MsgType:0100
BitMap:00100000001000000000010110000000001000001100000000000000000000110
003=309500
011=600007
022=022
024=0005
025=00
035=4444444444444444D16081012599368400000
041=70100094
042=7010000000000094
062=000012
063=TAGS
      1C=E0
```

6.2.2.4 Loyalty Balance Response Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	05	01	10	60	3C	01	80	2E	80	00	02	16
17	44	44	44	44	44	44	44	44	30	95	00	60	00	07	10	54
33	57	05	26	16	08	00	05	00	37	04	44	44	44	44	44	44
49	44	4D	16	08	10	12	59	93	68	40	00	00	30	30	30	30
65	37	31	31	30	35	35	33	36	30	30	30	30	30	30	30	30
81	37	30	31	30	30	30	39	34	00	16	00	14	35	33	30	30
97	30	30	30	30	30	30	32	35	33	31						

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	05H	01H	10H	`	<	01H	80	.	80	00H	02H	16H
17	D	D	D	D	D	D	D	D	0	95	00H	`	00H	07H	10H	T
33	W	05H	&	16H	08H	00H	05H	00H	7	04H	D	D	D	D	D	D
49	D	M	16H	08H	10H	12H	Y	93	h	@	00H	00H	0	0	0	0
65	7	1	1	0	5	5	3	6	0	0	0	0	0	0	0	0
81	7	0	1	0	0	0	9	4	00H	16H	00H	14H	5	3	0	0
97	0	0	0	0	0	0	2	5	3	1						

Parsed Message

TPDU: 6000000005
 MsgType:0110
 BitMap:0110000000011110000000000110000000001011101000000000000000000000010
 002=4444444444444444
 003=309500
 011=600007
 012=105457
 013=0526
 014=1608
 024=0005
 025=00
 035=0444444444444444D1608101259936840000
 037=000071105536
 038=000000
 039=00
 041=70100094
 063=TAGS
 53=000000002531

6.2.2.5 Loyalty Cash Request Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	05	00	00	02	00	30	20	05	80	20	C0	00	06	01
17	95	00	00	00	00	00	15	00	60	00	11	00	22	00	05	00
33	37	42	10	04	30	64	37	51	06	D1	60	81	01	25	99	36
49	84	00	00	0F	37	30	31	30	30	30	39	34	37	30	31	30
65	30	30	30	30	30	30	30	30	30	39	34	00	06	30	30	30
81	30	31	36	00	06	00	04	31	43	45	30					

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	05H	00H	00H	02H	00H	0	20H	05H	80	20H	C0	00H	06H	01H
17	95	00H	00H	00H	00H	00H	15H	00H	`	00H	11H	00H	"	00H	05H	00H
33	7	B	10H	04H	0	d	7	Q	06H	D1	`	81	01H	%	99	6
49	84	00H	00H	0FH	7	0	1	0	0	0	9	4	7	0	1	0
65	0	0	0	0	0	0	0	0	0	9	4	00H	06H	0	0	0
81	0	1	6	00H	06H	00H	04H	1	C	E	0					

Parsed Message

TPDU: 6000050000
 MsgType: 0200
 BitMap: 001100000010000000000101100000000010000011000000000000000000000110
 003=019500
 004=000000001500
 011=600011
 022=022
 024=0005
 025=00
 035=4444444444444444D16081012599368400000
 041=70100094
 042=7010000000000094
 062=000016
 063=TAGS
 1C=E0

6.2.2.6 Loyalty Cash Response Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	05	02	10	70	3C	01	80	0E	80	00	02	16
17	44	44	44	44	44	44	44	44	01	95	00	00	00	00	00	10
33	00	60	00	11	11	20	14	05	26	16	08	00	05	00	30	30
49	30	30	37	30	30	39	34	35	30	34	30	34	31	38	35	32
65	30	30	37	30	31	30	30	30	39	34	00	16	00	14	35	33
81	30	30	30	30	30	30	30	30	30	30	33	35				

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	05H	02H	10H	p	<	01H	80	0EH	80	00H	02H	16H
17	D	D	D	D	D	D	D	D	01H	95	00H	00H	00H	00H	00H	10H
33	00H	`	00H	11H	11H	20H	14H	05H	&	16H	08H	00H	05H	00H	0	0
49	0	0	7	0	0	9	4	5	0	4	0	4	1	8	5	2
65	0	0	7	0	1	0	0	0	9	4	00H	16H	00H	14H	5	3
81	0	0	0	0	0	0	0	0	0	0	3	5				

Parsed Message

TPDU: 6000000005
 MsgType:0210
 BitMap:011100000011110000000000110000000000011101000000000000000000000010
 002=4444444444444444
 003=019500
 004=000000001000
 011=600011
 012=112014
 013=0526
 014=1608
 024=0005
 025=00
 037=000070094504
 038=041852
 039=00
 041=70100094
 063=TAGS
 53=000000000035

6.3 Refund (Processing with Visa Argentina)

When processing refunds with Visa Argentina the terminal has to request two specific data from the original transaction to be refunded:

- Original Transaction Date: The date where the original transaction was processed at the terminal, format DDMMYY
- Original Transaction Invoice: The Invoice number (data sent in DE 62) of the original transaction.

Both elements are sent in DE 63 using the private element Tag 35 as additional prompts

6.3.1 Refund Request Sample Transaction

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	30	20	05	80	20	C0	00	06	20
17	00	00	00	00	00	00	26	00	60	00	27	00	22	00	03	00
33	33	54	13	33	00	89	02	01	02	D2	51	22	01	05	05	90
49	10	3F	30	30	30	30	30	30	30	31	30	30	30	30	30	30
65	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	32
81	34	00	24	00	22	33	35	30	38	36	32	31	32	33	34	35
97	36	30	38	31	32	30	31	30	35	31	36					

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	0	20H	05H	80	20H	C0	00H	06H	20H
17	00H	00H	00H	00H	00H	00H	&	00H	`	00H	'	00H	'''	00H	03H	00H
33	3	T	13H	3	00H	89	02H	01H	02H	D2	Q	'''	01H	05H	05H	90
49	10H	?	0	0	0	0	0	0	0	1	0	0	0	0	0	0
65	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	2
81	4	00H	\$	00H	'''	3	5	0	8	6	2	1	2	3	4	5
97	6	0	8	1	2	0	1	0	5	1	6					

Parsed Message

TPDU: 6000030000
MsgType:0200
BitMap:0011000000100000000001011000000000100000110000000000000000000000110
003=200000
004=000000002600
011=600027
022=022
024=0003
025=00
035=5413330089020102D2512201050590103
041=00000001
042=0000000000000001
062=000024
063=TAGS
35=08621234560812010516
--++*****--++*****
| | | | |
| | | | | +>Original Trans Date DDMMYY
| | | | | +> Prompt 2 ID **12**
| | | | | +>Add Prompt 2 Lengh **8**
| | | | |
| | | +>Original Invoice 6 digits
| | +> Prompt 1 ID **62**
| +>Add Prompt 1 Lengh **8**

6.3.2 Refund Response Sample Transaction

HEXADECIMAL REPRESENTATION

[illegible]

ASCII REPRESENTATION

[illegible]

Parsed Message

```
TPDU: 6000000003
MsgType:0210
BitMap:011000000011110000000000110000000000011101000000000000000000000000
002=5413330089020102
003=200000
011=600027
012=120842
013=0530
014=2512
024=0003
025=00
037=615112600027
038=084234
039=00
041=00000001
```

6.4 Other Transactions Sample

6.4.1 EMV Sale

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	30	38	05	80	20	C0	02	06	00
17	00	00	00	00	00	00	32	72	60	00	49	16	31	11	05	30
33	00	51	00	03	00	33	54	13	33	00	89	02	00	11	D2	51
49	26	01	07	93	60	80	5F	30	30	30	30	30	30	30	31	30
65	30	30	30	30	30	30	30	30	30	30	30	30	30	30	31	01
81	5F	2A	02	03	40	82	02	30	00	84	07	A0	00	00	00	04
97	10	10	5F	34	01	03	95	05	08	00	00	80	00	9A	03	16
113	05	30	9C	01	00	9F	02	06	00	00	00	00	32	72	9F	03
129	06	00	00	00	00	00	00	9F	09	02	00	02	9F	10	12	02
145	10	A5	00	0F	04	00	00	00	00	00	00	00	00	00	00	00
161	FF	9F	1A	02	05	91	9F	1E	08	41	4C	43	4C	4F	43	41
177	4C	9F	26	08	0E	0D	D0	31	90	C5	FD	26	9F	27	01	80
193	9F	33	03	E0	B8	C8	9F	34	03	41	03	02	9F	35	01	22
209	9F	36	02	00	27	9F	37	04	F0	93	33	95	9F	41	03	00
225	00	37	00	06	30	30	30	30	33	37	00	54	00	14	33	39
241	30	30	30	30	30	30	30	30	30	30	39	32	00	14	34	30
257	30	30	30	30	30	30	30	30	30	32	30	30	00	14	38	32
273	30	30	30	30	30	30	30	30	30	31	30	30	00	04	31	43
289	45	30														

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	0	8	05H	80	20H	C0	02H	06H	00H
17	00H	00H	00H	00H	00H	00H	2	r	`	00H	l	16H	1	11H	05H	0
33	00H	Q	00H	03H	00H	3	T	13H	3	00H	89	02H	00H	11H	D2	Q
49	&	01H	07H	93	`	80	_	0	0	0	0	0	0	0	1	0
65	0	0	0	0	0	0	0	0	0	0	0	0	0	1	01H	F
81	_	*	02H	03H	@	82	02H	0	00H	84	07H	A0	00H	00H	00H	04H
97	10H	10H	_	4	01H	03H	95	05H	08H	00H	00H	80	00H	9A	03H	16H
113	05H	0	9C	01H	00H	9F	02H	06H	00H	00H	00H	00H	2	r	9F	03H
129	06H	00H	00H	00H	00H	00H	00H	9F	09H	02H	00H	02H	9F	10H	12H	02H
145	10H	A5	00H	0FH	04H	00H	00H	00H	00H	00H	00H	00H	00H	00H	00H	00H
161	FFH	9F	1AH	02H	05H	91	9F	1EH	08H	A	L	C	L	O	C	A
177	L	9F	&	08H	0EH	0DH	D0	1	90	C5	FD	&	9F	'	01H	80
193	9F	3	03H	E0	B8	C8	9F	4	03H	A	03H	02H	9F	5	01H	''
209	9F	6	02H	00H	'	9F	7	04H	F0	93	3	95	9F	A	03H	00H
225	00H	7	00H	06H	0	0	0	0	3	7	00H	T	00H	14H	3	9
241	0	0	0	0	0	0	0	0	0	0	9	2	00H	14H	4	0
257	0	0	0	0	0	0	0	0	0	2	0	0	00H	14H	8	2
273	0	0	0	0	0	0	0	0	0	1	0	0	00H	04H	1	C
289	E	0														

Parsed Message

TPDU: 6000030000
MsgType: 0200
BitMap: 0011000000111000000001011000000000100000110000000000001000000110
003=000000
004=000000003272
011=600049
012=163111
013=0530
022=051
024=0003
025=00
035=5413330089020011D2512601079360805
041=00000001
042=0000000000000001
055=5F2A020340820230008407A00000000410105F340103950508000080009A031605309
C01009F02060000000032729F03060000000000009F090200029F10120210A5000F040000
000000000000000000FF9F1A0205919F1E08414C434C4F43414C9F26080E0DD03190C5FD2
69F2701809F3303E0B8C89F34034103029F3501229F360200279F3704F09333959F410300
0037
<EMVTAGS>
5F2A=0340
82=3000
84=A0000000041010
5F34=03
95=0800008000
9A=160530
9C=00
9F02=000000003272
9F03=000000000000
9F09=0002
9F10=0210A5000F0400000000000000000000FF
9F1A=0591
9F1E=414C434C4F43414C
9F26=0E0DD03190C5FD26
9F27=80
9F33=E0B8C8
9F34=410302
9F35=22
9F36=0027
9F37=F0933395
9F41=000037
</EMVTAGS>
062=000037
063=TAGS
 39=0000000000092
 40=000000000200
 82=000000000100
 1C=E0

6.4.2 Sale – Swiped

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	30	20	05	80	20	C0	00	06	00
17	00	00	00	00	00	00	17	38	60	00	39	00	22	00	03	00
33	33	54	13	33	00	89	02	01	02	D2	51	22	01	05	05	90
49	10	3F	30	30	30	30	30	30	30	31	30	30	30	30	30	30
65	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	33
81	30	00	54	00	14	33	39	30	30	30	30	30	30	30	30	30
97	31	33	38	00	14	34	30	30	30	30	30	30	30	30	30	30
113	31	30	30	00	14	38	32	30	30	30	30	30	30	30	30	30
129	31	35	30	00	04	31	43	45	30							

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	0	20H	05H	80	20H	C0	00H	06H	00H
17	00H	00H	00H	00H	00H	00H	17H	8	`	00H	9	00H	""	00H	03H	00H
33	3	T	13H	3	00H	89	02H	01H	02H	D2	Q	""	01H	05H	05H	90
49	10H	?	0	0	0	0	0	0	0	1	0	0	0	0	0	0
65	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	3
81	0	00H	T	00H	14H	3	9	0	0	0	0	0	0	0	0	0
97	1	3	8	00H	14H	4	0	0	0	0	0	0	0	0	0	0
113	1	0	0	00H	14H	8	2	0	0	0	0	0	0	0	0	0
129	1	5	0	00H	04H	1	C	E	0							

Parsed Message

TPDU: 6000030000
 MsgType: 0200
 BitMap: 001100000001000000000001011000000000010000011000000000000000000000000000110
 003=000000
 004=000000001738
 011=600039
 022=022
 024=0003
 025=00
 035=5413330089020102D2512201050590103
 041=00000001
 042=0000000000000001
 062=000030
 063=TAGS
 39=0000000000138
 40=0000000000100
 82=0000000000150
 1C=E0

6.4.3 Sale Reversal

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	04	00	70	24	05	80	00	C0	00	06	16
17	54	13	33	00	89	02	01	02	00	00	00	00	00	00	00	24
33	34	60	00	41	25	12	00	22	00	03	00	30	30	30	30	30
49	30	30	31	30	30	30	30	30	30	30	30	30	30	30	30	30
65	30	31	00	06	30	30	30	30	33	31	00	54	00	14	33	39
81	30	30	30	30	30	30	30	30	30	31	38	34	00	14	34	30
97	30	30	30	30	30	30	30	30	30	32	35	30	00	14	38	32
113	30	30	30	30	30	30	30	30	30	32	30	30	00	04	31	43
129	45	30														

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	04H	00H	p	\$	05H	80	00H	C0	00H	06H	16H
17	T	13H	3	00H	89	02H	01H	02H	00H	00H	00H	00H	00H	00H	00H	\$
33	4	`	00H	A	%	12H	00H	'''	00H	03H	00H	0	0	0	0	0
49	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
65	0	1	00H	06H	0	0	0	0	3	1	00H	T	00H	14H	3	9
81	0	0	0	0	0	0	0	0	0	1	8	4	00H	14H	4	0
97	0	0	0	0	0	0	0	0	0	2	5	0	00H	14H	8	2
113	0	0	0	0	0	0	0	0	0	2	0	0	00H	04H	1	C
129	E	0														

Parsed Message

TPDU: 6000030000
 MsgType:0400
 BitMap:0111000000100100000001011000000000000000110000000000000000000000110
 002=5413330089020102
 003=000000
 004=000000002434
 011=600041
 014=2512
 022=022
 024=0003
 025=00
 041=00000001
 042=0000000000000001
 062=000031
 063=TAGS
 39=000000000184
 40=000000000250
 82=000000000200
 1C=E0

6.4.4 Sale Void

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	70	3C	05	80	0C	C0	00	06	16
17	54	13	33	00	89	02	01	02	02	00	00	00	00	00	00	20
33	38	60	00	54	09	26	07	05	31	25	12	00	22	00	03	00
49	36	31	35	32	30	39	36	30	30	30	35	33	32	36	30	37
65	38	34	30	30	30	30	30	30	30	31	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	33
97	39	00	54	00	14	33	39	30	30	30	30	30	30	30	30	30
113	31	33	38	00	14	34	30	30	30	30	30	30	30	30	30	30
129	31	30	30	00	14	38	32	30	30	30	30	30	30	30	30	30
145	31	35	30	00	04	31	43	45	30							

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	p	<	05H	80	0C H	C0	00H	06H	16H
17	T	13H	3	00H	89	02H	01H	02H	02H	00H	00H	00H	00H	00H	00H	20H
33	8	`	00H	T	09H	&	07H	05H	1	%	12H	00H	"	00H	03H	00H
49	6	1	5	2	0	9	6	0	0	0	5	3	2	6	0	7
65	8	4	0	0	0	0	0	0	0	1	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	3
97	9	00H	T	00H	14H	3	9	0	0	0	0	0	0	0	0	0
113	1	3	8	00H	14H	4	0	0	0	0	0	0	0	0	0	0
129	1	0	0	00H	14H	8	2	0	0	0	0	0	0	0	0	0
145	1	5	0	00H	04H	1	C	E	0							

Parsed Message

TPDU: 6000030000
MsgType: 0200
BitMap: 0111000000111100000000101100000000000011001100000000000000000000110
002=5413330089020102
003=020000
004=000000002038
011=600054
012=092607
013=0531
014=2512
022=022
024=0003
025=00
037=615209600053
038=260784
041=00000001
042=0000000000000001
062=000039
063=TAGS
 39=0000000000138
 40=0000000000100
 82=0000000000150
 1C=E0

6.4.5 Cash Advance

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	30	20	05	80	20	C0	00	06	01
17	00	00	00	00	00	00	21	00	60	00	42	00	22	00	03	00
33	33	54	13	33	00	89	02	01	02	D2	51	22	01	05	05	90
49	10	3F	30	30	30	30	30	30	30	31	30	30	30	30	30	30
65	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	33
81	32	00	06	00	04	31	43	45	30							

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	0	20H	05H	80	20H	C0	00H	06H	01H
17	00H	00H	00H	00H	00H	00H	!	00H	`	00H	B	00H	'''	00H	03H	00H
33	3	T	13H	3	00H	89	02H	01H	02H	D2	Q	'''	01H	05H	05H	90
49	10H	?	0	0	0	0	0	0	0	1	0	0	0	0	0	0
65	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	3
81	2	00H	06H	00H	04H	1	C	E	0							

Parsed Message

```
TPDU: 6000030000
MsgType:0200
BitMap:00110000001000000000001011000000000100000110000000000000000000110
003=010000
004=0000000002100
011=600042
022=022
024=0003
025=00
035=5413330089020102D2512201050590103
041=00000001
042=0000000000000001
062=000032
063=1C=E0
```

6.4.6 Cash Advance Void

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	70	3C	05	80	0C	C0	00	06	16
17	54	13	33	00	89	02	01	02	03	00	00	00	00	00	00	21
33	00	60	00	56	09	26	32	05	31	25	12	00	22	00	03	00
49	36	31	35	32	30	39	36	30	30	30	35	35	32	36	33	32
65	35	33	30	30	30	30	30	30	30	31	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	34
97	30	00	06	00	04	31	43	45	30							

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	p	<	05H	80	0CH	C0	00H	06H	16H
17	T	13H	3	00H	89	02H	01H	02H	03H	00H	00H	00H	00H	00H	00H	!
33	00H	`	00H	V	09H	&	2	05H	1	%	12H	00H	"	00H	03H	00H
49	6	1	5	2	0	9	6	0	0	0	5	5	2	6	3	2
65	5	3	0	0	0	0	0	0	0	1	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	4
97	0	00H	06H	00H	04H	1	C	E	0							

Parsed Message

TPDU: 6000030000
 MsgType: 0200
 BitMap: 0111000000111100000000101100000000000011001100000000000000000000110
 002=5413330089020102
 003=030000
 004=000000002100
 011=600056
 012=092632
 013=0531
 014=2512
 022=022
 024=0003
 025=00
 037=615209600055
 038=263253
 041=00000001
 042=0000000000000001
 062=000040
 063=TAGS
 1C=E0

6.4.7 Payment

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	30	20	05	80	20	C0	00	06	21
17	00	00	00	00	00	00	25	00	60	00	44	00	22	00	03	00
33	33	54	13	33	00	89	02	01	02	D2	51	22	01	05	05	90
49	10	3F	30	30	30	30	30	30	30	31	30	30	30	30	30	30
65	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	33
81	33	00	06	00	04	31	43	45	30							

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	0	20H	05H	80	20H	C0	00H	06H	!
17	00H	00H	00H	00H	00H	00H	%	00H	`	00H	D	00H	""	00H	03H	00H
33	3	T	13H	3	00H	89	02H	01H	02H	D2	Q	""	01H	05H	05H	90
49	10H	?	0	0	0	0	0	0	0	1	0	0	0	0	0	0
65	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	3
81	3	00H	06H	00H	04H	1	C	E	0							

Parsed Message

```
TPDU: 6000030000
MsgType:0200
BitMap:001100000010000000000010110000000001000001100000000000000000000110
003=210000
004=0000000002500
011=600044
022=022
024=0003
025=00
035=5413330089020102D2512201050590103
041=00000001
042=0000000000000001
062=000033
063=TAGS
1C=E0
```

6.4.8 Payment Void

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	02	00	70	3C	05	80	0C	C0	00	06	16
17	54	13	33	00	89	02	01	02	23	00	00	00	00	00	00	25
33	00	60	00	58	09	38	03	05	31	25	12	00	22	00	03	00
49	36	31	35	32	30	39	36	30	30	30	35	37	33	38	30	33
65	39	38	30	30	30	30	30	30	30	31	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	00	06	30	30	30	30	34
97	31	00	06	00	04	31	43	45	30							

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	02H	00H	p	<	05H	80	0CH	C0	00H	06H	16H
17	T	13H	3	00H	89	02H	01H	02H	#	00H	00H	00H	00H	00H	00H	%
33	00H	`	00H	X	09H	8	03H	05H	1	%	12H	00H	"	00H	03H	00H
49	6	1	5	2	0	9	6	0	0	0	5	7	3	8	0	3
65	9	8	0	0	0	0	0	0	0	1	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	00H	06H	0	0	0	0	4
97	1	00H	06H	00H	04H	1	C	E	0							

Parsed Message

TPDU: 6000030000
 MsgType:0200
 BitMap:0111000000111100000000101100000000000011001100000000000000000000000000110
 002=5413330089020102
 003=230000
 004=000000002500
 011=600058
 012=093803
 013=0531
 014=2512
 022=022
 024=0003
 025=00
 037=615209600057
 038=380398
 041=00000001
 042=0000000000000001
 062=000041
 063=TAGS
 1C=E0

6.4.9 Settlement

6.4.9.1 Settlement Header Request

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	05	00	20	20	01	00	00	C0	00	12	92
17	00	00	60	00	65	00	03	30	30	30	30	30	30	30	31	30
33	30	30	30	30	30	30	30	30	30	30	30	30	30	31	00	06
49	30	30	30	30	30	33	02	10	30	30	32	30	30	30	30	30
65	30	30	30	34	33	38	34	30	30	30	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	30	30	30	30	30	30	30
97	30	31	35	30	31	30	30	30	30	30	30	30	30	30	30	30
113	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
129	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
145	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
161	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
177	30	30	31	30	30	30	30	30	30	30	30	30	35	31	35	30
193	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
209	31	30	30	30	30	30	30	30	30	32	36	32	31	30	30	30
225	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
241	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
257	30	30	30	30	30	30	30	30	30	30						

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	05H	00H	20H	20H	01H	00H	00H	C0	00H	12H	92
17	00H	00H	`	00H	e	00H	03H	0	0	0	0	0	0	0	1	0
33	0	0	0	0	0	0	0	0	0	0	0	0	0	1	00H	06H
49	0	0	0	0	0	3	02H	10H	0	0	2	0	0	0	0	0
65	0	0	0	4	3	8	4	0	0	0	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
97	0	1	5	0	1	0	0	0	0	0	0	0	0	0	0	0
113	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
145	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
161	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
177	0	0	1	0	0	0	0	0	0	0	0	0	5	1	5	0
193	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
209	1	0	0	0	0	0	0	0	0	2	6	2	1	0	0	0
225	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
241	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
257	0	0	0	0	0	0	0	0	0	0						

Parsed Message

[illegible]

Settlement Totals on DE 63 Parsing

Pos	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45				
Value	0	0	2	0	0	0	0	0	0	0	0	4	3	8	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	5	0	1						
Field	Sales Count	Sales Total														Refund Count	Refund Amount										Cash Count	Cash Amount																					
	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90				
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Debit Sale Count	Debit Sales Amount														Debit Ref. Count	Debit Refund Amount										Auth Sale Count	Auth Sale Amount																						
	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135				
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	5	1	5						
Auth Ref Count	Auth Refund Amount														HYP RSVD	HYPERCOM RESERVED										Void Sales Count	Void Sales Amount																						
	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180				
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	2	6	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0				
Void Ref Count	Void Refund Amount														Payments Count	Payments Amount										Void Pay Count	Void Payments Amount																						
	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210																			
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																			
Loyalty Count	Loyalty Amount														CashBack Count	CashBack Amount																																	

6.4.9.2 Settlement Header Response (Batch Upload)

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	00	00	03	05	10	20	38	01	80	0A	80	00	02	92
17	00	00	60	00	65	09	43	28	05	31	00	03	00	36	31	35
33	32	30	39	36	30	30	30	36	35	39	35	30	30	30	30	30
49	30	30	31	00	31	50	52	4F	43	45	53	41	4E	44	4F	20
65	52	45	4D	49	53	49	4F	4E	20	45	4C	45	43	54	52	4F
81	4E	49	43	41												

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	00H	00H	03H	05H	10H	20H	8	01H	80	0AH	80	00H	02H	92
17	00H	00H	`	00H	e	09H	C	(	05H	1	00H	03H	00H	6	1	5
33	2	0	9	6	0	0	0	6	5	9	5	0	0	0	0	0
49	0	0	1	00H	1	P	R	O	C	E	S	A	N	D	O	20H
65	R	E	M	I	S	I	O	N	20H	E	L	E	C	T	R	O
81	N	I	C	A												

Parsed Message

TPDU: 6000000003
 MsgType: 0510
 BitMap: 001000000001110000000000011000000000000101010000000000000000000000010
 003=920000
 011=600065
 012=094328
 013=0531
 024=0003
 025=00
 037=615209600065
 039=95
 041=00000001
 063=PROCESANDO REMISION ELECTRONICA

6.4.9.3 Batch Upload

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	03	20	70	3C	05	80	0C	C0	00	16	16
17	54	13	33	00	89	02	01	02	00	00	01	00	00	00	00	22
33	39	60	00	66	09	41	17	05	31	25	12	00	22	00	03	00
49	36	31	35	32	30	39	36	30	30	30	35	39	34	31	31	37
65	38	35	30	30	30	30	30	30	30	31	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	00	22	30	32	30	30	36
97	30	30	30	35	39	20	20	20	20	20	20	20	20	20	20	20
113	20	00	06	30	30	30	30	34	32	00	54	00	14	33	39	30
129	30	30	30	30	30	30	30	30	31	33	38	00	14	34	30	30
145	30	30	30	30	30	30	30	30	31	30	31	00	14	38	32	30
161	30	30	30	30	30	30	30	30	31	35	30	00	04	31	43	45
177	30															

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	03H	20H	p	<	05H	80	0CH	C0	00H	16H	16H
17	T	13H	3	00H	89	02H	01H	02H	00H	00H	01H	00H	00H	00H	00H	""
33	9	`	00H	f	09H	A	17H	05H	1	%	12H	00H	""	00H	03H	00H
49	6	1	5	2	0	9	6	0	0	0	5	9	4	1	1	7
65	8	5	0	0	0	0	0	0	0	1	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	00H	""	0	2	0	0	6
97	0	0	0	5	9	20H	20H	20H	20H	20H	20H	20H	20H	20H	20H	20H
113	20H	00H	06H	0	0	0	0	4	2	00H	T	00H	14H	3	9	0
129	0	0	0	0	0	0	0	0	1	3	8	00H	14H	4	0	0
145	0	0	0	0	0	0	0	0	1	0	1	00H	14H	8	2	0
161	0	0	0	0	0	0	0	0	1	5	0	00H	04H	1	C	E
177	0															

Parsed Message

TPDU: 6000030000
MsgType: 0320
BitMap: 011100000011110000000010110000000000001100110000000000000000000010110
002=5413330089020102
003=000001
004=000000002239
011=600066
012=094117
013=0531
014=2512
022=022
024=0003
025=00
037=615209600059
038=411785
041=00000001
042=0000000000000001
060=0200600059
062=000042
063=TAGS
 39=0000000000138
 40=0000000000101
 82=0000000000150
 1C=E0

6.4.9.4 Batch Upload Last Trans in Batch

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	03	20	70	3C	05	80	0C	C0	00	16	16
17	54	13	33	00	89	02	01	02	21	00	00	00	00	00	00	26
33	21	60	00	70	09	42	41	05	31	25	12	00	22	00	03	00
49	36	31	35	32	30	39	36	30	30	30	36	34	34	32	34	31
65	37	38	30	30	30	30	30	30	30	31	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	00	22	30	32	30	30	36
97	30	30	30	36	34	20	20	20	20	20	20	20	20	20	20	20
113	20	00	06	30	30	30	30	34	36	00	06	00	04	31	43	45
129	30															

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	03H	20H	p	<	05H	80	0CH	C0	00H	16H	16H
17	T	13H	3	00H	89	02H	01H	02H	!	00H	00H	00H	00H	00H	00H	&
33	!	`	00H	p	09H	B	A	05H	1	%	12H	00H	'''	00H	03H	00H
49	6	1	5	2	0	9	6	0	0	0	6	4	4	2	4	1
65	7	8	0	0	0	0	0	0	0	1	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	00H	'''	0	2	0	0	6
97	0	0	0	6	4	20H	20H	20H	20H	20H	20H	20H	20H	20H	20H	20H
113	20H	00H	06H	0	0	0	0	4	6	00H	06H	00H	04H	1	C	E
129	0															

Parsed Message

TPDU: 6000030000
MsgType: 0320
BitMap: 011100000011110000000010110000000000001100110000000000000000000010110
002=5413330089020102
003=210000
004=000000002621
011=600070
012=094241
013=0531
014=2512
022=022
024=0003
025=00
037=615209600064
038=424178
041=00000001
042=0000000000000001
060=0200600064
062=000046
063=TAGS
1C=E0

6.4.9.5 Settlement Trailer

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	05	00	20	20	01	00	00	C0	00	12	96
17	00	00	60	00	71	00	03	30	30	30	30	30	30	30	31	30
33	30	30	30	30	30	30	30	30	30	30	30	30	30	31	00	06
49	30	30	30	30	30	33	02	10	30	30	32	30	30	30	30	30
65	30	30	30	34	33	38	34	30	30	30	30	30	30	30	30	30
81	30	30	30	30	30	30	30	30	31	30	30	30	30	30	30	30
97	30	31	35	30	31	30	30	30	30	30	30	30	30	30	30	30
113	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
129	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
145	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
161	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
177	30	30	31	30	30	30	30	30	30	30	30	30	35	31	35	30
193	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
209	31	30	30	30	30	30	30	30	30	32	36	32	31	30	30	30
225	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
241	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30
257	30	30	30	30	30	30	30	30	30	30						

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	`	00H	03H	00H	00H	05H	00H	20H	20H	01H	00H	00H	C0	00H	12H	96
17	00H	00H	`	00H	q	00H	03H	0	0	0	0	0	0	0	1	0
33	0	0	0	0	0	0	0	0	0	0	0	0	0	1	00H	06H
49	0	0	0	0	0	3	02H	10H	0	0	2	0	0	0	0	0
65	0	0	0	4	3	8	4	0	0	0	0	0	0	0	0	0
81	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
97	0	1	5	0	1	0	0	0	0	0	0	0	0	0	0	0
113	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
129	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
145	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
161	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
177	0	0	1	0	0	0	0	0	0	0	0	0	5	1	5	0
193	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
209	1	0	0	0	0	0	0	0	0	2	6	2	1	0	0	0
225	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
241	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
257	0	0	0	0	0	0	0	0	0	0						

Parsed Message

```
TPDU: 6000030000  
MsgType:0500  
BitMap:00100000000100000000000000100000000000000001100000000000000000010010  
003=960000  
011=600071  
024=0003  
041=00000001  
042=00000000000000001  
060=000003  
063=002000000004384000000000000000000010000000015010000000000000000000000000  
00000000000000000000000000000000000000000000000000000000000000000000000010000000005150000000  
000000000000100000000026210000000000000000000000000000000000000000000000000000
```

6.4.10 Echo Test

HEXADECIMAL REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	60	00	03	00	00	08	00	20	00	01	00	00	80	00	00	99
17	00	00	00	03	30	30	30	30	30	30	30	31				

ASCII REPRESENTATION

POS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		00H	03H	00H	00H	08H	00H	20H	00H	01H	00H	00H	80	00H	00H	99
17	00H	00H	00H	03H	0	0	0	0	0	0	0	1				

Parsed Message

TPDU: 6000030000

MsgType:0800

```
Bitmap:001000000000000000000001000000000000001000000000000000
```

003=990000

024=0003

$$041 = 00000001$$

7.0 TCP/IP Transport Considerations

7.1 ISO 8583 over TCP/IP

The process of transmitting ISO 8583 transactions over TCP/IP is accomplished preceding each transaction with two bytes representing the length of the transaction data to be transmitted.

One or more transactions could be transmitted sequentially over the same socket.

One or more LLL+Transaction Occurrences						
LLLL	1st ISO Transaction	LLLL	2nd ISO Transaction		LLLL	n th ISO Transaction

7.1.1 Length calculation

LLLL means that two binary bytes are going to be used to give the length of the data being transmitted.

Let's use this Echo Test transactions as example

```

      1           2           3           4           5
POS 12345678901234567890123456789012345678901234567890123456
001C60000300000800200001000080000099000000033030303030303031
*****
|                               |
|                               | +> Variable Length Transaction Data
+> LLLL
```

We have 56 hexadecimal characters, but remember this is a hexadecimal representation, so actually only 28 bytes are really used to transmit.

So let's decode the first two bytes

```
00 Hex -> 00 Decimal
1C Hex -> 28 Decimal
```

The first byte must be multiplied by 256 and then we add the second byte,
(00 * 256) + 28 = 28 -> bytes

In reverse if we are calculating the length of a transmitted transaction

The first byte is given by the formula *Integer*(Length Div 256)

The second byte given by the formula (Length Modulus 256)

So if we have a 280 byte transaction

1st Byte = 280 / 256 = 1.09375 -> we use the integer part = 1 represented as Hex "01"

2nd Byte = 280 mod 256 = 24 -> represented as Hex "18"

So if we put bytes 1st and 2nd together represented as Hex we have "0118", that effectively represented as a decimal gives the magic number "280" which is the Length of the transaction we are going to transmit .

7.2 Using SSL/TLS Sockets

“ISSwitch Server” secures TCP/IP communications using TLS / SSL sockets. SSL components support SSL v3, TLS 1.0, 1.1 and 1.2 versions of this protocol.

SSL Server component can use self-signed certificates as well-known certification brands issued ones.

The minimum key size used must be 2048 bits, and the hash algorithm is SHA-1 with RSA encryption.

The general process for establishing a secure TCP/IP socket is depicted in the following illustration.

